AbosCluster

User Manual

Version 1.0

Clustering of weight trajectories
after bariatric surgery

Sparse longitudinal data, latent class models and functional k-means

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Reader's guide: the data analyzed

All numerical results and figures in this manual are derived from a single dataset drawn from the ABOS cohort (*A Biological Atlas of Severe Obesity*, CHU de Lille). **Appendix A** describes it in detail.

— **Source sample**: 1 443 patients who underwent bariatric surgery, contributing 8 734 measurements, i.e. 2 to 7 visits per patient (6 on average), spread over a maximum follow-up of 95 months. Visit times are irregular and patient-specific (sparse longitudinal data).

— **Analysis sample**: smoking status is missing for 5 patients (0.3%), across all 26 of their measurements. These rows are **removed before any analysis**, leaving **1 438 patients and 8 708 measurements**. The descriptive plots (Section 3.5) and all clustering results (Sections 5.10, 6.4 and 8.3) refer to this sample. No outlying values of weight or TWL were removed (Appendix, Section A.2).

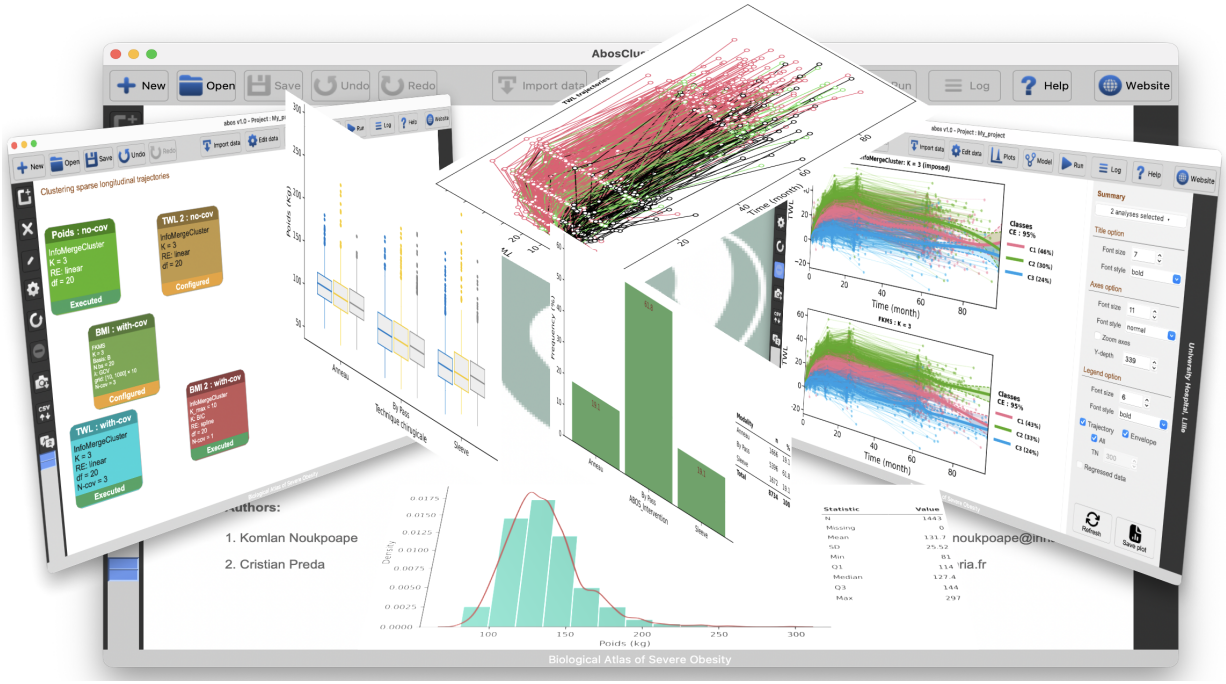
— **Weight outcomes studied**: body weight (kg; variable `Poids`), body mass index (BMI) and percentage total weight loss (TWL). All three are transformations of body weight (Appendix, Section A.3).

— **Covariates**: age at surgery, surgical procedure, smoking status and diabetes status. A *covariate-adjusted* analysis includes all four; an *unadjusted* analysis includes none.

— **Labels in the figures**: category labels, legend titles and axis titles were entered in French when the figures were produced (e.g. *Anneau Gastrique*, *Poids (kg)*); they are reproduced verbatim in the parameter windows shown below, and their English equivalents are given in the figure captions.

Where to find the examples. The descriptive plots are in Section 3.5, the InfoMergeCluster analysis in Section 5.10 (Figure 11), the FKMS analysis in Section 6.4 (Figure 12), the comparison of raw and covariate-adjusted (regressed) data in Figure 13, and the comparison of partitions using the adjusted Rand index (ARI) and the Jaccard index in Section 8.3.

With its user-friendly interface, AbosCluster performs, without any programming, exploratory analyses (Section 3.5) and clustering (Section 4) of sparse longitudinal data observed at irregular time points. The screenshot below gives an illustration. Predefined dialog boxes allow several clustering models (with or without covariates) to be specified and compared by means of agreement indices.



Contents

Foreword **5**

1 AbosCluster, a standalone executable **6**

1.1 What “executable” means . . . 6
1.2 Launching the application . . . 6
1.3 Window layout . . . 7
1.4 Projects . . . 7
1.5 Language, font and keyboard . . . 7
1.6 Data structure . . . 8

2 Button reference **8**

2.1 Top toolbar . . . 9
2.2 Left sidebar . . . 10
2.3 Right sidebar . . . 11
2.4 Command availability . . . 11
2.5 Execution panel and interface lock . . . 11

3 Preparing and describing the data **12**

3.1 Importing the data . . . 12
3.2 Preparing the data (Edit data) . . . 12
3.3 Descriptive summary (Overview stat) . . . 13
3.4 Exporting data and tables . . . 13
3.5 Descriptive plots (Plots) . . . 14
3.5.1 Worked examples . . . 14
3.6 Options common to all plots . . . 19

4 Clustering trajectories **19**

4.1 The clinical question and its challenges . . . 19
4.2 Overview . . . 20
4.3 Step 1: create an analysis box . . . 20
4.4 Step 2: choose and configure the method . . . 20
4.5 Step 3: run . . . 20
4.6 Reading an analysis box . . . 21
4.7 Managing boxes . . . 21

5 InfoMergeCluster: a mixture of linear mixed models **21**

5.1 Model . . . 22
5.2 Mean trajectories: penalized splines . . . 22
5.3 Individual variability (re) . . . 23
5.4 Prior distributions on the variance structure . . . 23
5.5 Initialization by micro-clusters and merging . . . 23
5.6 Choosing the number of classes . . . 24
5.7 Covariates . . . 24
5.8 Assignment of patients . . . 24
5.9 Parameters . . . 25
5.10 Clustering with InfoMergeCluster: a worked example . . . 25

6 FKMS: functional k-means **28**
6.1 Principle . . . 28
6.2 Smoothing and selection of λ by generalized cross-validation . . . 28
6.3 Covariates, confidence envelopes and number of classes . . . 29
6.4 Clustering with FKMS: a worked example . . . 30

7 InfoMergeCluster vs FKMS **33**

8 Reading, comparing and exporting results **34**
8.1 The results panel . . . 34
8.2 Plot-specific options . . . 34
8.3 Comparing several analyses . . . 36
8.3.1 Comparison boxes . . . 36
8.3.2 Agreement indices . . . 37
8.3.3 Displaying and exporting the indices . . . 38
8.3.4 Example: three analyses of body weight . . . 38
8.4 Exporting . . . 39

A Description of the ABOS cohort data **41**
A.1 Data and observation scheme . . . 41
A.2 Missing values . . . 42
A.3 Deterministic relationships between the response variables . . . 43

References **44**

Foreword

Weight response to bariatric surgery varies considerably between patients. After a phase of rapid weight loss, some patients stabilize, others regain weight after reaching their nadir, and others never achieve sufficient weight loss. Identifying these response profiles, and the patients who follow each of them, supports postoperative follow-up, the search for prognostic factors and the pathophysiological understanding of the response. AbosCluster performs this clustering without programming, from the repeated weight measurements (body weight, body mass index or percentage total weight loss, TWL) collected at follow-up visits.

Intended audience. Bariatric surgeons, endocrinologists, dietitians, physicians working in specialized obesity centers, clinical research associates and biostatisticians. The manual assumes familiarity with standard weight-outcome measures and descriptive statistics; linear mixed models, latent classes and splines are introduced where they are first used.

Organization. The manual follows the order in which a study is conducted.

Section	Content	Question addressed
1	The software and the project	How do I launch AbosCluster, organize a study and structure the data?
2	Buttons	What does each command do, and when is it available?
3	Data preparation and description	Are the data complete, consistent and fit for analysis?
4	Steps of a clustering analysis	How do I create, configure and run an analysis?
5	InfoMergeCluster	What does the probabilistic method do, and how should its parameters be set?
6	FKMS	What does functional k-means do, and how should its parameters be set?
7	InfoMergeCluster vs FKMS	What distinguishes the two methods?
8	Results	How do I read, interpret and export the clusters obtained, and quantify the agreement between two partitions (ARI, Jaccard index)?

Conventions.

— Interface labels are set in bold and kept in English, as they appear on screen: **Run**, **Edit data**, **Overview stat**.

— A sequence of clicks is indicated by an arrow: **Edit data** → **Overview stat** → **Export stat**.

— The *response* is the weight measure whose course over time is studied (body weight, body mass index or percentage total weight loss, TWL). *Time* denotes the time elapsed since surgery. *Covariates* are the patient characteristics included in the model (age, diabetes status, smoking status, surgical procedure, etc.).

— A *cluster* (or *class*) is a group of patients sharing the same pattern of weight change; the *mean trajectory* of a cluster is the estimated curve summarizing that pattern.

— Blue boxes: statistical meaning of a result or setting. Brown boxes: clinical cautions and caveats for interpretation.

Status of the tool

AbosCluster is a tool for exploratory analysis and clinical research. Its clusters are statistical constructs derived from a sample; they are neither diagnoses nor treatment indications. Any decision regarding an individual patient remains a matter of clinical judgment.

1. AbosCluster, a standalone executable

AbosCluster v1.0 is a standalone desktop application developed in Python within the *Spyder* environment, using the **tkinter** module. It starts with a double-click and requires no installation of Python or of any library. Its name refers to ABOS (*A Biological Atlas of Severe Obesity*), a prospective cohort sponsored by Lille University Hospital (CHU de Lille) and registered as NCT01129297. The **Website** button opens the study record on ClinicalTrials.gov.

1.1 What "executable" means

The program is written in Python and then "frozen": the interpreter, libraries and graphical resources are bundled into a single package. For a given version, dataset and set of parameters, results are identical provided the random seed is fixed (`seed`, Sections 5.9 and 6.3); if the seed is left empty, it changes at every run. This reproducibility holds only *within AbosCluster*: the same seed does not reproduce the random draws obtained in R (`set.seed()`), whose random number generator, sampling functions and some initialization algorithms differ (see the note in Section 5.9).

Bundled component	Role in AbosCluster
Tkinter and tkmacosx	Graphical user interface: windows, buttons, lists, dialog boxes
pandas and NumPy	Reading, filtering and computations on data tables
InfoMergeCluster	Latent class linear mixed model (Section 5)
FKMS	Functional k-means (Section 6)
Matplotlib	All plots
python-docx and openpyxl	Word and Excel exports
`AbosCluster` folder	Button icons and user manual

1.2 Launching the application

1. Copy the application folder to the local disk, without separating the executable from its resources.
2. Double-click the AbosCluster executable. On macOS, the first launch may require authorization in the security settings.
3. The main window opens; its title bar displays "AbosCluster v1.0", followed by the name of the open project.

1.3 Window layout

Area	Position	Content
Top toolbar	Top	Project, data, plots, analyses, execution, results, help
Sidebar	Left	Management of analysis boxes, stop, screenshot, export, language, size and zoom slider
Central panel	Center	Workspace: tables, plots, analysis and comparison boxes, dialog boxes
Sidebar	Right	Management of model comparison boxes, size slider for these boxes
Bottom bar	Bottom	Study title: *Biological Atlas of Severe Obesity*

Each button of the top toolbar displays its own panel in the central area. Depending on the panel displayed, the vertical slider of the left sidebar sets either the size of the analysis boxes or the zoom level of the selected plot; the slider of the right sidebar sets the size of the model comparison boxes.

1.4 Projects

All work is carried out within a *project*. Until a project has been created (**New**) or opened (**Open**), the analysis commands are unavailable. A project consists of two items stored side by side:

— a human-readable file `<name>.json` describing the study: analysis and comparison boxes, methods, parameters, comparison indices, filters, variable conversions and plot settings;

— a folder `<name>_fichiers` containing `donnees.pkl`: the imported and edited data and the results of the executed analyses.

Safeguards:

— **Autosave**: the project is saved every 5 minutes if it has been modified.

— **History**: the last 20 actions (import, editing, filtering, creation, configuration, execution, deletion, etc.) can be undone (**Undo**) or redone (**Redo**).

— **Protected closing**: if the project contains unsaved changes, closing it offers **Save**, **Don't save** or **Cancel**. The same prompt appears before another project is created or opened.

Health data

A project folder contains individual health data: store it in accordance with institutional rules (restricted access, backup), using pseudonymized identifiers. Since `donnees.pkl` is a Python serialization (pickle) file, open only projects of known provenance.

1.5 Language, font and keyboard

The language button  in the left sidebar opens the *Langue et police / Language and font* dialog box, in which the language of the instructions (*English* or *French*), the font and the font size are selected. The choice takes effect immediately, and the language is remembered. Button labels, parameter names and plot text remain in English. The Enter key confirms a dialog box; the Esc key closes it.

1.6 Data structure

AbosCluster reads tables in CSV, Excel (`.xlsx`, `.xls`) or text (`.txt`) format. The data must be in *long format*: **one row per visit**, not one row per patient. The first row contains the column names.

Patient	Time	Poids	Age	Diabetes	Surgery
P001	3.0	109.1	42.1	no	sleeve
P001	6.2	99.5	42.1	no	sleeve
P001	12.1	90.2	42.1	no	sleeve
P002	1.1	108.9	55.7	yes	RYGB
P002	11.8	88.9	55.7	yes	RYGB

Table 1 – Expected long format: identifier, postoperative time (months), response (body weight, kg), covariates.

— **Identifier**: one code per patient, identical across all of that patient's visits (column `Patient` by default).

— **Time**: time elapsed since surgery, expressed in a single unit (months are recommended). Visits need not be synchronized across patients, nor regularly spaced.

— **Response**: one numeric value per visit.

— **Covariates**: optional, either numeric (age) or categorical (diabetes status, smoking status, etc.); a categorical variable is automatically encoded as indicator (dummy) variables.

2. Button reference

Commands are distributed between three permanent bars and the buttons specific to each panel. An unavailable command is grayed out.

At startup, the central panel presents the software and its authors (Figure 1); the five areas are those described in Section 1.3. The first action is to create a project (**New**) or open an existing one (**Open**).

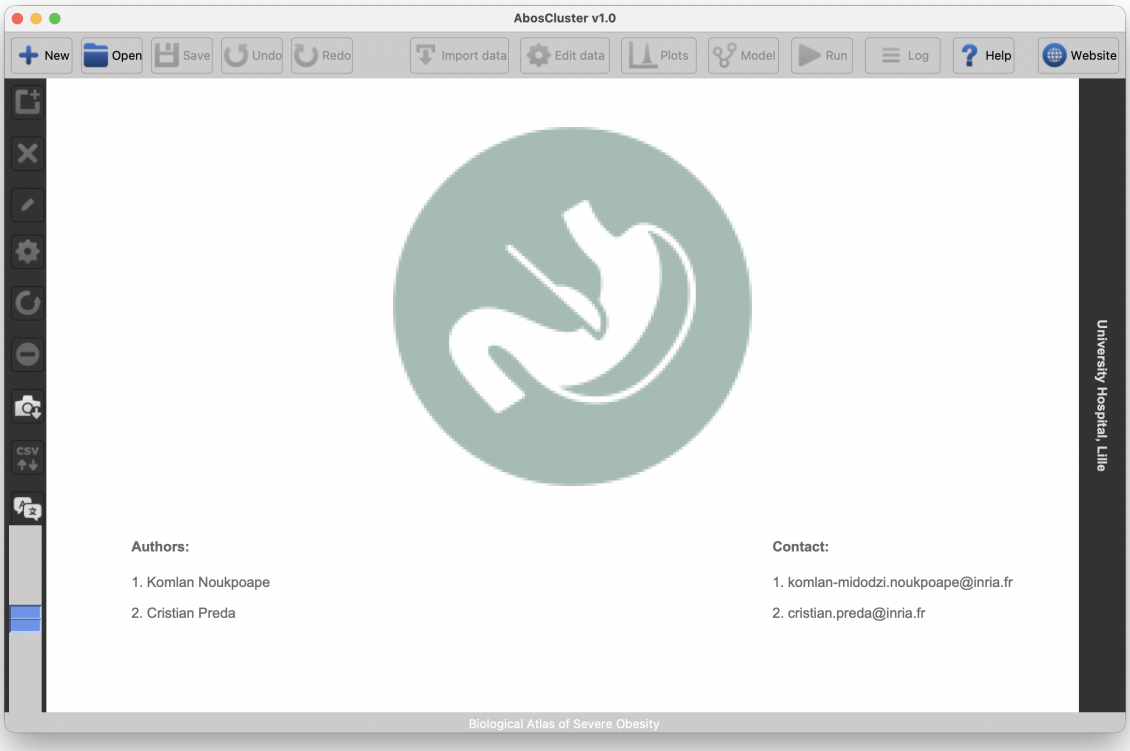


Figure 1 – AbosCluster interface at startup, before a project is created or opened. Only **New**, **Open**, **Help** and **Website**, together with the interface screenshot and language selection, are active; all other commands remain grayed out.

2.1 Top toolbar



Figure 2 – AbosCluster top toolbar, from left to right.

Button	Function	When to use it
New	Creates a project: name, then location of the JSON file	At the start of any study
Open	Opens a project and restores its data, boxes, parameters and results	To resume a study
Save	Saves the open project	After any important step
Undo / Redo	Undoes or redoes an action (20 levels)	After a handling error
Import data	Imports one or more tables (CSV, Excel, text)	Immediately after creating the project
Edit data	Opens the data preparation panel (Section 3.2)	Before any analysis
Plots	Opens the descriptive plots (Section 3.5)	Data exploration
Model	Displays the analysis box panel	Creation and configuration (Section 4)
Run	Runs all configured analyses	After configuration

Button	Function	When to use it
Log	Displays the results of the executed analyses (Section 8)	After execution
Help	Opens this manual	To look up an explanation
Website	Opens the ABOS study record (NCT01129297)	For background on the cohort

2.2 Left sidebar

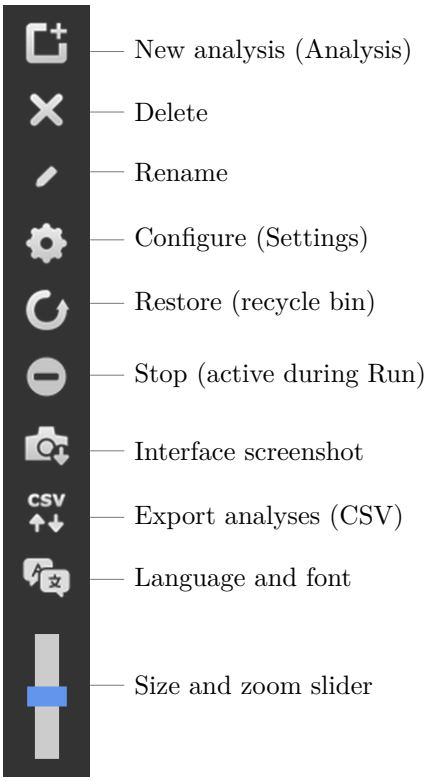


Figure 3 – Left sidebar. Light icons are shown on a dark background, as in the software.

Button	Function
New analysis	Creates an analysis box: name, then optional color
Delete	Moves the selected analysis box to the recycle bin (recoverable)
Rename	Renames the selected analysis box; its parameters, results and membership in comparison boxes follow the new name
Configure	Opens the method selection and its parameters (equivalent to double-clicking the box)
Restore	Opens the recycle bin: restoration or permanent deletion
Stop	Interrupts the running analyses; active (blue background) only during an execution
Screenshot	Saves a PNG image of the window as currently displayed
Export	Exports the list of analyses and the cluster assigned to each patient (Section 8.4)
Language	Selection of the instruction language, font and font size
Vertical slider	Size of the analysis boxes (analysis panel) or zoom level of the selected plot

Left-sidebar commands act only on analysis boxes. Model comparison boxes have their own commands, located on the right sidebar.

2.3 Right sidebar

The right sidebar manages the *model comparison boxes*, which gather executed analyses in order to quantify the agreement between their partitions (Section 8.3). Its icons are the same as those of the left sidebar, in the same order.

Button	Function
New comparison	Creates a comparison box: name, then optional color (default color if the selection is cancelled)
Delete	Moves the selected comparison box to its recycle bin; the analyses it contains are left unchanged
Rename	Renames the selected comparison box
Configure	Opens the comparison dialog box: list of models, indices (ARI, Jaccard), color (equivalent to double-clicking the box)
Restore	Opens the comparison recycle bin: restoration or permanent deletion
Vertical slider	Size of the comparison boxes, independent of that of the analysis boxes (80% by default)

2.4 Command availability

No import is possible outside a project, and no analysis without data:

Condition	Active commands
Always	**New**, **Open**, **Help**, **Website**, screenshot, language
A project is open	**Import data**
A project is open and data have been imported	**Save**, **Edit data**, **Plots**, **Model**, **Run**, **Log**; left sidebar: new analysis, delete, rename, configure, recycle bin, export; right sidebar: comparison box commands
An action can be undone or redone	**Undo**, **Redo**
An execution is in progress	Stop *only*: all other commands are locked (Section 2.5)

2.5 Execution panel and interface lock

During **Run**, a blue panel in the center of the workspace shows the analysis in progress (for example “Running: TWL-IM (2 / 4)”), the elapsed time and the estimated remaining time. The estimate is based on the duration of the most recent computation of each analysis or, failing that, on the mean duration of the analyses already completed. The panel can be moved with the mouse and disappears when the computations are finished.

During the computations the interface is *locked*: buttons, boxes, lists, fields and sliders ignore mouse clicks and keystrokes, and the pointer turns into a busy cursor. Only the *Stop* button (blue background) and the panel remain active. The mouse wheel still scrolls, except over lists, numeric fields and sliders. The lock, which prevents any modification while estimation is under way, is released at the end of the computations or after an interruption.

3. Preparing and describing the data

A unit error, an incorrectly computed time variable or a duplicated identifier is enough to produce a spurious cluster. Data checking and preparation must therefore precede any modeling.

3.1 Importing the data

Import data accepts several files; if they share the same columns (for example, extracts by center or by enrollment period), they are stacked. For an Excel workbook, the sheets to be imported are selected and stacked in the same way. A message confirms that the data have been loaded or reports the cause of failure.

3.2 Preparing the data (Edit data)

Edit data opens the preparation panel. At the top are three blocks: on the left, the check boxes **Missing value imputation**, **Convert variable** and **Filtering data**, each of which opens its own dialog box; in the center, search-and-replace of category values and column renaming; on the right, the columns and rows to be displayed. Four buttons are available (Figure 4): **View & Edit** applies the settings and displays the table, **Overview stat** produces the descriptive summary (Section 3.3), **Export stat** exports it, and **Export clean** exports the prepared data (Section 3.4).



Figure 4 – Buttons of the **Edit data** panel.

Operation	Settings	Effect
Display / Refresh (**View & Edit**)	Columns and number of rows to display (**Parameters for overview**)	Applies the saved settings and displays the table, with the numbers of patients, rows and columns
Impute (**Missing value imputation**)	Dialog box: **Column**, then **Quantitative** method and **Qualitative** method	Handles the missing values of the selected column, or of all columns (details below)
Transform (**Convert variable**)	**Categorizing variables**, **Convert to double**	Treats a numerically coded variable as categorical (e.g. 0/1), or converts numeric text to a real number
Filter (**Filtering data**)	Minimum and maximum (quantitative variables); categories to retain (categorical variables)	Restricts the analysis to a subpopulation or to a follow-up window
Search and replace (**Search & change modality**)	Column, value searched for, replacement value	Harmonizes inconsistent coding (e.g. “Sleeve”, “sleeve”)
Rename (**Rename column**)	Old and new name	Clarifies a column name

Missing values. The **Missing value imputation** check box opens a three-row dialog box.

— **Column**: the column to be processed, or **All columns**. Each column is listed with its type and its number of missing values, recomputed every time the dialog box opens, for example `M000..TabacFumeur (character, 26 NA)`.

— **Quantitative**: `mean` or `median` replaces the missing value with the mean or the median; `delete` removes the row.

— **Qualitative**: `modality reference` replaces the missing value with the mode; `delete` removes the row.

With **All columns**, each method is applied to every column of the corresponding type. Saved choices (**Save**) take effect at the next **View & Edit**; unticking the check box cancels them. Any numeric column not declared categorical through **Categorizing variables** is treated as quantitative.

Imputation and longitudinal data

AbosCluster's methods accommodate incomplete and irregular follow-up: a patient seen at 3, 12 and 36 months contributes as is. Do *not* impute the response at missed visits: an imputed mean or median erases individual variability and may create an artificial "stable" cluster. Imputation should be confined to covariates; if incomplete rows are rare, deleting them is the safer option. In the ABOS data analyzed here, only smoking status is missing, for five patients (Appendix, Section A.2): this column is best handled with the `delete` option rather than by assigning these patients an unobserved status.

3.3 Descriptive summary (Overview stat)

Edit data → **Overview stat** produces a summary of each column, computed on the filtered data.

Table column	Meaning
Characteristics	Variable name
Type	`double`, `integer` (quantitative), `character` or `categorical` (qualitative)
Unique	Number of categories of a categorical variable
Modality	Name of each category
Missing	Number of missing values
N	Number of non-missing values
mean (SD) / (%)	Mean (standard deviation) of a quantitative variable; percentage of a category

Visits or patients?

In long format, these statistics are computed over *visits*: a patient seen ten times carries ten times the weight. For a time-invariant characteristic (age, sex), counts should be made per patient, for instance with the `patients` option of bar plots (Section 3.5).

3.4 Exporting data and tables

Export clean exports the cleaned and filtered data, exactly as they will be analyzed. **Export stat** exports the descriptive summary to Excel (with formatting preserved), Word or CSV.

3.5 Descriptive plots (Plots)

In the **Plots** panel, tick a plot type in the **Graphiques** list, then a variable under **Response variables**, adjust the parameters and click **Plot**. Several variables of the same plot type are drawn one below the other.

Plot	What it shows	Main parameters
Trajectory	Individual trajectories of the response as a function of time since surgery	Title, axis labels, grouping variable, interval breaks (*breaks*), labels, legend title, color, line width
Box plot	Distribution of a variable, optionally by intervals of an X variable	X variable, breaks and labels for the intervals of X, grouping variable
Bar plot	Frequencies of the categories of a variable	Axis labels, breaks and labels for a quantitative variable, color, counting per row (`rows`) or per patient (`patients`)
Histogram	Distribution of a quantitative variable	Number of bins (Sturges' rule by default), scale: count, frequency (%) or density

Breaks are separated by commas (`-Inf, 40, 50, Inf`) and labels by semicolons (`< 40; 40-50; >= 50`). Intervals are left-closed, $[a, b)$, for the grouping variable of trajectory plots, and right-closed, $(a, b]$, for box plots and bar plots.

Worked examples

The four plots below are based on the analysis sample: 1 438 patients and 8 708 measurements, after removal of the 5 patients with missing smoking status (Appendix, Section A.2).

Each diagram numbers the successive clicks; step 4 gives the values entered. A field that is not listed keeps its default value; a field left empty takes the automatic value indicated in the diagram.

Individual trajectories (Trajectory).

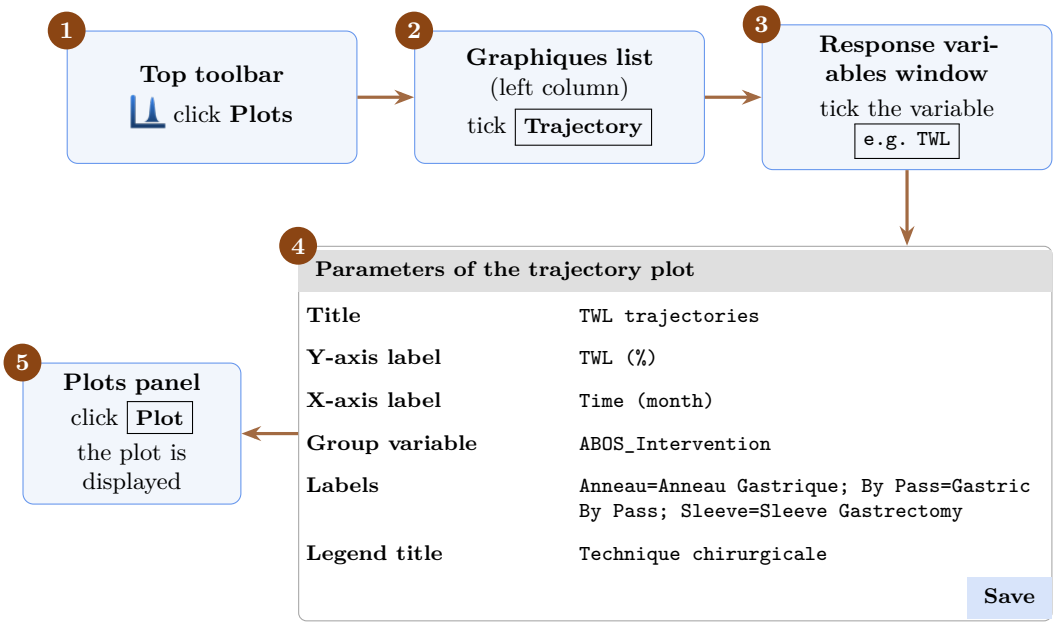


Figure 5 shows each patient's TWL as a function of time since surgery, with one line per patient, colored by surgical procedure.

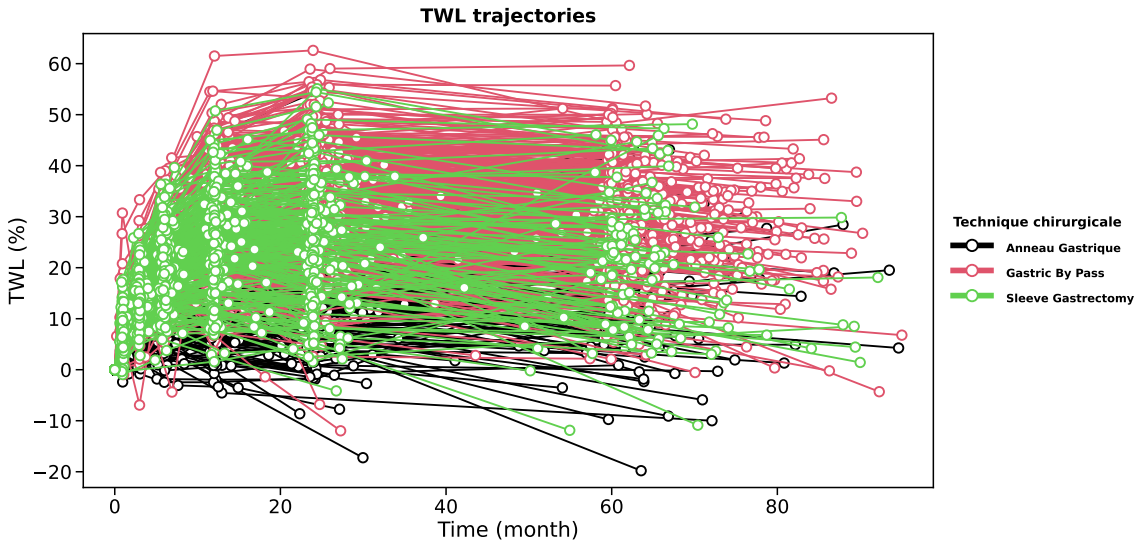


Figure 5 – Individual TWL (%) trajectories by surgical procedure (Anneau Gastrique = gastric banding, Gastric By Pass = gastric bypass, Sleeve Gastrectomy = sleeve gastrectomy).

Measurements are concentrated around the scheduled visits (3, 6, 12, 24 and 60 months), within a few weeks. Between 24 and 60 months they are less frequent (Table 12), and beyond the 60-month visit they become sparser still owing to loss to follow-up. Gastric bypass trajectories lie generally above those of gastric banding, with sleeve gastrectomy in between; this is a qualitative, unadjusted reading. Negative values indicate a weight above the preoperative weight. To reduce overplotting, use the **TN** option (Section 8.2) or filter by procedure.

Box plots (Box plot).

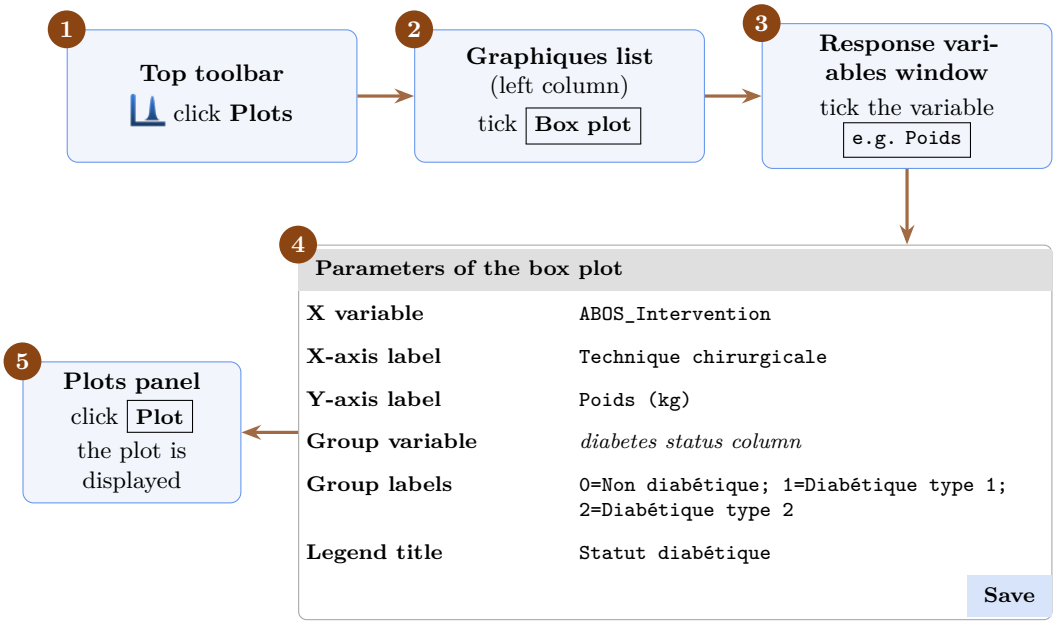


Figure 6 shows body weight by surgical procedure (X variable) and diabetes status (grouping variable). **Group labels** replaces the codes 0, 1 and 2 with labels and sets the order of the

legend. In the right-hand column, the **Mean point** block (displayed for box plots only) adds the mean of each distribution: tick **View point**, then choose the color (**Color**: `red` by default) and the point size (**Point size**: 6 by default); the plot is redrawn immediately.

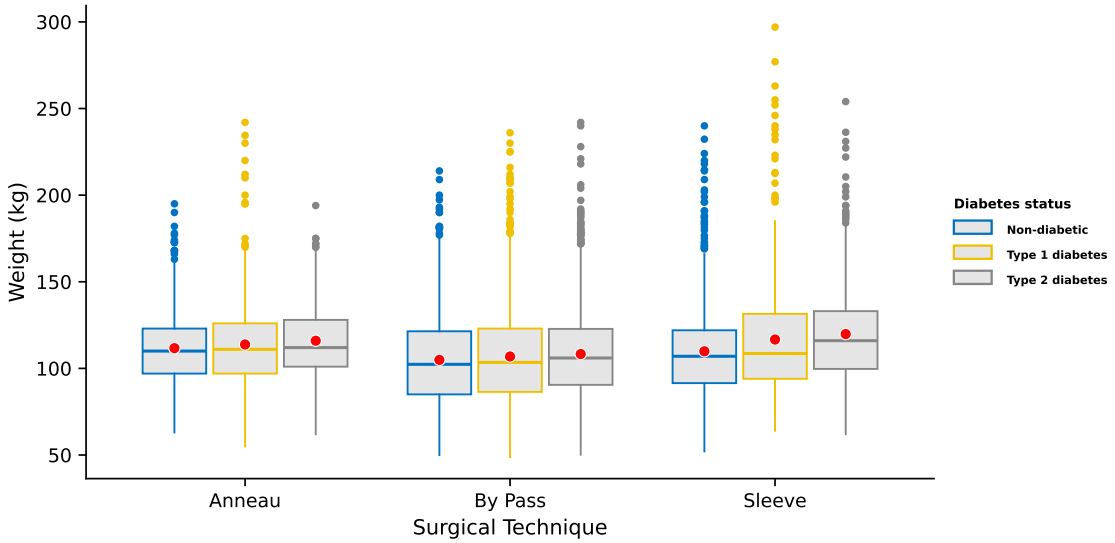


Figure 6 – Distribution of body weight (kg) by surgical procedure (Anneau = gastric banding, By Pass = gastric bypass, Sleeve = sleeve gastrectomy) and diabetes status (Non diabétique = no diabetes, Diabétique type 1 = type 1 diabetes, Diabétique type 2 = type 2 diabetes). Red dots: means (**Mean point** block).

The box spans the first to the third quartile and the inner line marks the median; the whiskers extend to the most extreme observations lying within 1.5 times the interquartile range of the box edges, and points beyond them, drawn in the color of the box, are outliers. The red dot marks the mean.

Medians are similar, roughly between 100 and 115 kg. The median is slightly higher in patients with type 2 diabetes, particularly after sleeve gastrectomy, and central values are slightly lower for gastric bypass. In the sleeve gastrectomy group, patients with type 1 diabetes are more dispersed and include the sample maximum, close to 300 kg. These differences remain small relative to within-group dispersion.

In every box, the mean (red dot) lies a few kilograms above the median: high values pull the mean upward, a sign of right skewness (see also the histogram, Figure 8). The median is therefore a better summary of typical weight in each group; the gap between mean and median, largest in patients with type 1 diabetes who underwent sleeve gastrectomy, identifies the groups most influenced by extreme values.

In severe obesity a long right tail is expected: an outlier is not necessarily an error, although the most extreme values should be verified. More importantly, each box pools pre- and postoperative measurements from patients at different stages of follow-up. Comparing procedures or diabetes status requires a fixed time point (preoperative weight, weight at 12 months), which can be obtained by filtering (Section 3.2). Counts refer to measurements, not to patients.

Bar plot (Bar plot).

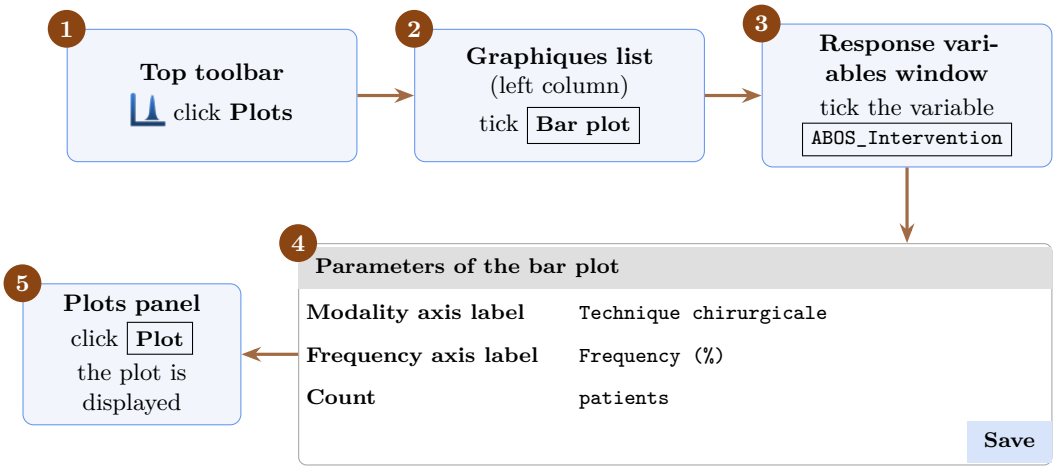
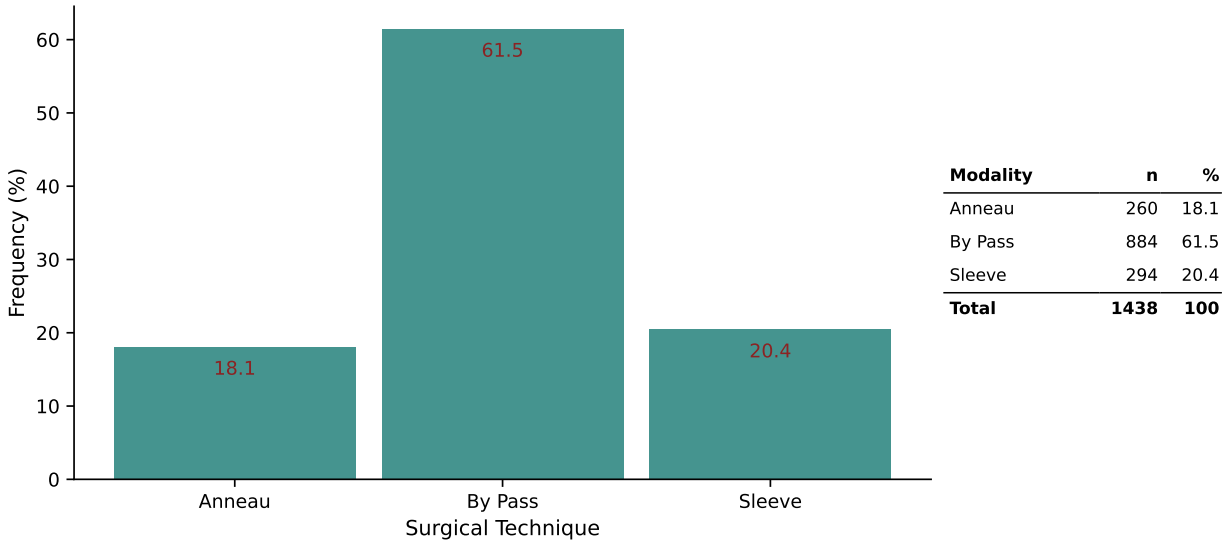


Figure 7 shows the distribution of patients across the three surgical procedures.



Modality	n	%
Anneau	260	18.1
By Pass	884	61.5
Sleeve	294	20.4
Total	**1438**	**100**

Figure 7 – Distribution of surgical procedures (Anneau = gastric banding, By Pass = gastric bypass, Sleeve = sleeve gastrectomy), counted per patient (`patients` option).

With the `patients` option, each patient is counted once (first non-missing value). Of the 1 438 patients, 884 underwent gastric bypass (61.5%), 294 sleeve gastrectomy (20.4%) and 260 gastric banding (18.1%).

For a time-invariant characteristic, this patient-level count is the one that describes the population. Counting per measurement (`rows` option) would yield 61.9% for gastric bypass, 19.1% for sleeve gastrectomy and 18.9% for gastric banding, out of 8 708 measurements. The discrepancy reflects the mean number of measurements per patient: about 6.3 (banding), 6.1 (bypass) and 5.7 (sleeve). Patients who underwent sleeve gastrectomy are thus followed for a shorter period or less frequently, which can be checked from the distribution of follow-up times by procedure.

Histogram (Histogram).

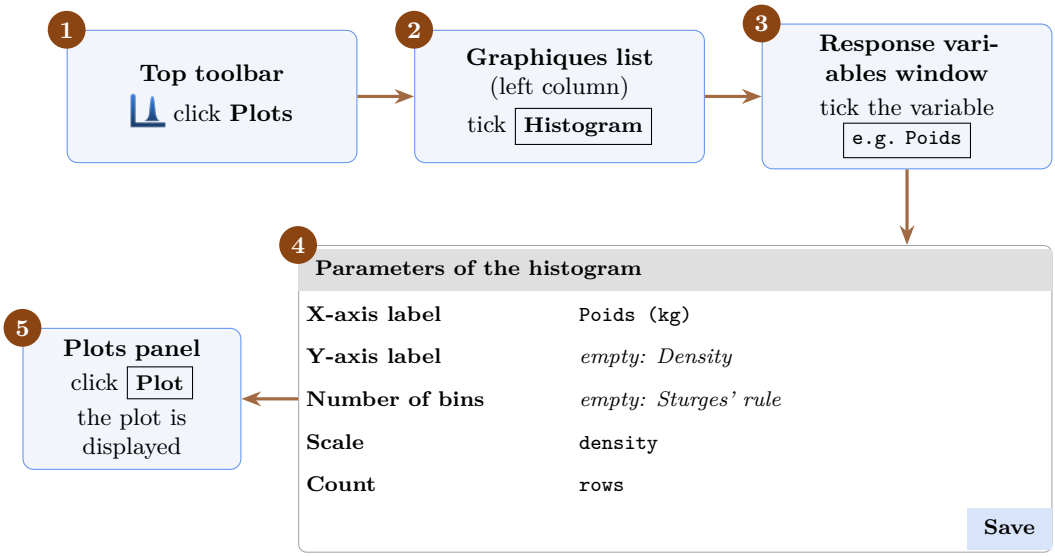
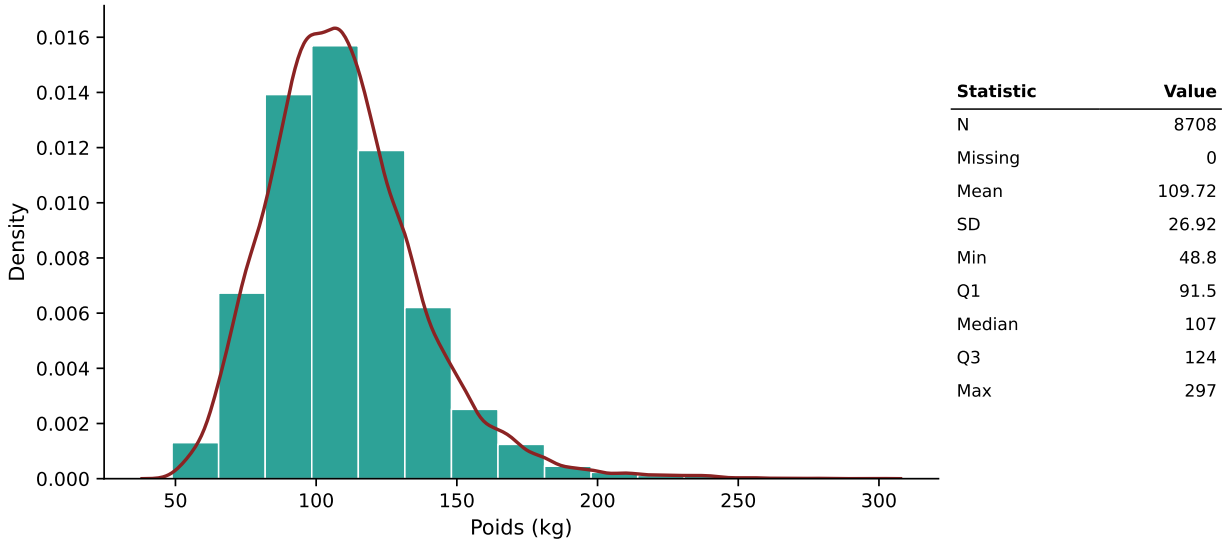


Figure 8 shows the distribution of body weight on the density scale, together with the kernel density estimate and descriptive statistics.



Statistic	Value
N	8708
Missing	0
Mean	109.72
SD	26.92
Min	48.8
Q1	91.5
Median	107
Q3	124
Max	297

Figure 8 – Distribution of body weight (kg) in the analysis sample (8 708 measurements from 1 438 patients, `rows` option): density histogram, estimated density and descriptive statistics.

The distribution is unimodal and right-skewed: mean 109.7 kg, median 107 kg, over the 8 708 measurements of the analysis sample. The median and interquartile range (91.5 to 124 kg) summarize it better than the mean and standard deviation (26.9 kg). Values range from 48.8 to 297 kg; the maximum is the value close to 300 kg already identified on the box plots (Figure 6). In the source sample (8 734 measurements, Table 13), the mean was 109.8 kg and the first quartile 91.6 kg: excluding the 26 measurements with missing smoking status leaves the distribution essentially unchanged. Because all visits are pooled, these figures describe weight over the course of follow-up, not body size at a given time point.

Descriptive plots and inference

These plots describe the sample without performing any test: they are not sufficient to conclude that groups differ. Their purpose is to check the data and to guide the choice of response, covariates and follow-up window before clustering.

3.6 Options common to all plots

Settings in the right-hand column (none of them triggers a recomputation):

Setting	Effect
Title option, **Axes option**, **Legend option**	Font size and style of titles, axes and legends
Mean point	Box plots only: **View point** displays the mean of each distribution; **Color** and **Point size** set the color and size of the points
Zoom axes	**Normal** (automatic limits) or limits entered under **X** and **Y** (minimum and maximum of the horizontal and vertical axes)
Y-depth	Height, in pixels, of each plot; common to all plots
Plot selection	Clicking a plot selects it (bluish background); clicking outside the plots clears the selection
Vertical slider, dragging	Zooming and panning of the selected plot; a double-click restores the original view
Hovering	The coordinates of the nearest curve point are displayed, together with the axis titles
Legend	Clicking a class shows only its elements; a second click shows all elements again
Refresh	Redraws the plot with the current settings
Save plot	Saves the selected plots as PNG, PDF, SVG, JPEG or TIFF files, or in a Word document, at the chosen resolution

Axis limits apply to the selected plot, or to all plots if none is selected. Plots that are to be compared should share the same axis limits.

4. Clustering trajectories

4.1 The clinical question and its challenges

Two questions arise: *how many* distinct patterns of weight change does the population contain, and *to which* pattern does each patient belong? Bariatric follow-up data make this task difficult:

— visits are **sparse**: few measurements per patient (at most seven in the ABOS cohort), spread over several years;

— they are **irregular**: follow-up times differ from one patient to another;

— follow-up **thins out**: the proportion of patients still being seen decreases over time;

— **intra-individual variability** (fluctuations, measurement error) is superimposed on the differences between profiles.

Conventional clustering of measurements taken “at 12 months” or “at 24 months” would require synchronized visits and would exclude many patients. AbosCluster instead treats each trajectory as a continuous function of time, estimated from the available visits whatever their timing.

4.2 Overview

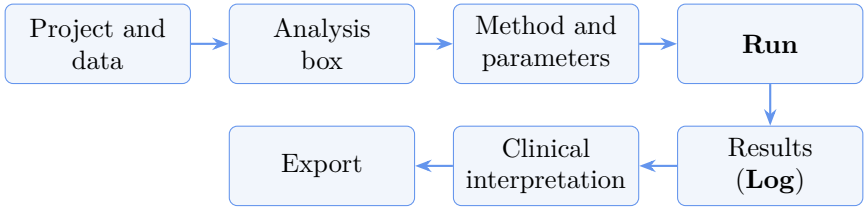


Figure 9 – Steps of a clustering analysis in AbosCluster. Several analyses can follow this workflow in parallel within the same project, so that they can be compared.

4.3 Step 1: create an analysis box

1. Click **Model**, then *New analysis* in the left sidebar.
2. Enter an explicit name (for example “Poids”) and, optionally, choose a color.
3. The box appears in the *Clustering sparse longitudinal trajectories* panel. It can be moved with the mouse; the vertical slider of the left sidebar sets the size of all analysis boxes.

It is advisable to combine the response and the method in the name (“Poids-IM”, “TWL-FK”): followed by `no-cov` or `with-cov`, the name serves as the title of the result plots (Section 8).

4.4 Step 2: choose and configure the method

1. Select the box (click), then click *Configure*, or double-click the box.
2. Select the method from the list: **InfoMergeCluster** (Section 5) or **FKMS** (Section 6).
3. The parameter dialog box opens (and is reopened in the same way). Its four sections, *Data*, *Model*, *Algorithm* and *Plot*, scroll with the mouse wheel; the handle at the bottom right corner enlarges the window.
4. Under *Data*, select the identifier (**Identifier**), the follow-up time (**Time**) and the weight response (**Response**), and tick any covariates (**Covariates**).
5. Confirm with **Save**. The configuration is stored and will be executed at the next **Run**.

Changing a model parameter discards the computed result; changing a *Plot* option only updates the plot.

4.5 Step 3: run

Run executes all configured analyses one after the other; boxes that have not been configured are skipped and flagged. The interface is locked during the computations (Section 2.5). Results are displayed at the end and remain accessible through **Log**. *Stop* cancels the pending analyses and the one in progress; completed analyses keep their results.

4.6 Reading an analysis box

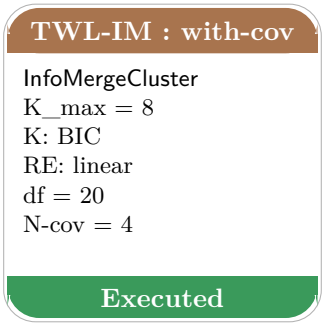


Figure 10 – An analysis box after configuration and execution.

Status	Meaning
Created	Box created; method not yet selected or not configured
Configured	Parameters saved; the analysis will be executed at the next **Run**
Waiting	Queued for execution
Running	Computation in progress
Executed	Result available for the current configuration

To quantify the agreement between the partitions of several executed analyses, group them in a comparison box (Section 8.3; example in Section 8.3.4).

4.7 Managing boxes

Delete moves the box, including its configuration and result, to the recycle bin; *Restore* retrieves it or deletes it permanently. *Rename* changes the name of a box without losing its parameters or its result. All these actions can be undone with **Undo**.

Single and multiple selection. Click: selects a box (blue outline). Shift, Ctrl or Cmd + click: adds a box to, or removes it from, the selection. Click on the background: clears the selection. Multiple selection is used to place several analyses in a comparison box (Section 8.3). Left-sidebar commands (configure, delete, rename) act on the last box clicked.

5. InfoMergeCluster: a mixture of linear mixed models

InfoMergeCluster is a *probabilistic* method: the population is modeled as a mixture of K *latent classes*, each with its own mean trajectory, from which every patient deviates in an individual way. The method jointly estimates the trajectories, the mixing proportions and the variance components, and selects K.

5.1 Model

Let y_{ij} denote the weight response (body weight, TWL or BMI) of patient i ($i = 1, \dots, N$) at their j-th visit, at time t_{ij} ($j = 1, \dots, n_i$). Conditionally on patient i belonging to class k:

$$y_{ij} = \underbrace{\mathbf{B}(t_{ij})^\top \boldsymbol{\beta}_k}_{\text{mean trajectory}} + \underbrace{\mathbf{x}_{ij}^\top \boldsymbol{\gamma}_k}_{\text{covariates}} + \underbrace{\mathbf{z}(t_{ij})^\top \mathbf{u}_i}_{\text{random effect}} + \underbrace{\varepsilon_{ij}}_{\text{residual error}}, \tag{1}$$

$$\mathbf{u}_i \sim \mathcal{N}_q(\mathbf{0}, \mathbf{G}), \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

Term	Clinical interpretation
$\mathbf{B}(t)^\top \boldsymbol{\beta}_k$	Mean trajectory of class k: a linear combination of cubic B-spline functions of follow-up time, flexible enough to capture rapid weight loss, an extremum (nadir or peak), a plateau or weight regain
$\mathbf{x}_{ij}^\top \boldsymbol{\gamma}_k$	Covariate effects, specific to each class (covariates are centered, so that the mean trajectory corresponds to a patient with average characteristics)
$\mathbf{z}(t)^\top \mathbf{u}_i$	Patient-specific random effect: persistent deviation from the class trajectory (level, slope, etc.), with covariance matrix $\mathbf{G}$ common to all classes
ε_{ij}	Residual error: transient fluctuation and measurement error, with variance σ^2
π_k	Proportion of the population belonging to class k (mixing proportion)

The likelihood contribution of each patient is a finite mixture (McLachlan and Peel, 2000) of multivariate normal densities arising from linear mixed models (Laird and Ware, 1982):

$$f(\mathbf{y}_i) = \sum_{k=1}^{K} \pi_k \, \varphi\Big(\mathbf{y}_i;\ \mathbf{X}_i \boldsymbol{\theta}_k,\ \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^\top + \sigma^2 \mathbf{I}_{n_i}\Big), \tag{2}$$

where $\mathbf{X}_i$ and $\mathbf{Z}_i$ stack the row vectors $(\mathbf{B}(t_{ij})^\top, \mathbf{x}_{ij}^\top)$ and $\mathbf{z}(t_{ij})^\top$ over the n_i visits, and $\boldsymbol{\theta}_k = (\boldsymbol{\beta}_k, \boldsymbol{\gamma}_k)$. A patient with only two measurements contributes in proportion to the information they provide.

Why a mixed model?

Without the term $\mathbf{z}(t)^\top \mathbf{u}_i$, two patients in the same class would differ only by noise that is independent from one visit to the next. A patient who consistently loses three points more than the rest of their class shows a *persistent* deviation: the mixed model attributes it to individual variability rather than creating an additional class.

5.2 Mean trajectories: penalized splines

The basis comprises `df` B-spline functions (de Boor, 2001) of degree `degree` (15 and 3 by default), with interior knots placed at quantiles of the observed times (so that the basis is finer where visits are dense). Curvature is penalized so that the curves do not reproduce noise (O'Sullivan, 1986; Ruppert et al., 2003) (`penalty`):

$$\text{pen}(\boldsymbol{\beta}_k) = \lambda \, \boldsymbol{\beta}_k^\top \mathbf{P} \, \boldsymbol{\beta}_k, \qquad P_{jl} = \int B_j^{(m)}(u) \, B_l^{(m)}(u) \, \mathrm{d}u, \tag{3}$$

where $m =$ `pen_order` is the order of the penalized derivative ($m = 2$: curvature). The smoothing parameter λ is updated at each iteration (Fellner–Schall update (Wood and Fasiolo, 2017)); the complexity of each curve is measured by its effective degrees of freedom (*edf*), which are smaller than `df`. Results depend little on `df`, provided it is large enough.

5.3 Individual variability (re)

`re`	**Random effect**	**Clinical reading**
`intercept`	u_{i0}	The patient lies persistently above or below their class, by a constant offset
`linear` (default)	$u_{i0} + u_{i1}t$	The deviation may also widen or narrow over time
`spline`	combination of basis functions	The random effect may take any shape

5.4 Prior distributions on the variance structure

To separate the random effect (covariance $\mathbf{G}$) from the residual error (variance σ^2) despite the small number of visits per patient, InfoMergeCluster uses weakly informative prior distributions, *calibrated on the data* through the empirical variogram (Diggle, 1988) of the within-patient residuals: an *inverse-Wishart* prior ($\mathcal{IW}$) for the matrix $\mathbf{G}$ and an *inverse-gamma* prior ($\mathcal{IG}$) for the variance σ^2:

$$\mathbf{G} \sim \mathcal{IW}(\nu_0, \mathbf{\Psi}_0), \qquad \sigma^2 \sim \mathcal{IG}(a, b), \quad b = (a+1)\,\tau^2, \tag{4}$$

where the nugget effect τ^2 (the limiting semivariance between two very closely spaced visits) sets the mode of σ^2, and the remaining part of the variance (sill minus nugget) calibrates the scale matrix $\mathbf{\Psi}_0$ of $\mathbf{G}$. The parameters are estimated by their posterior mode, computed with a penalized EM algorithm (Dempster et al., 1977).

5.5 Initialization by micro-clusters and merging

The EM algorithm converges to a local maximum that depends on the starting partition. Rather than multiplying random starts for each value of K, InfoMergeCluster builds the candidate partitions through successive merges:

1. **Micro-clusters.** The individual trajectories, summarized by their coefficients on the spline basis, are partitioned by k-means into K_0 micro-clusters, with $K_0 > K_{\max}$ (`K0`; empty: automatic choice). This is the only random step of the algorithm (`nstart` initializations, seed `seed`).
2. **Merges.** At each step, the two classes whose merger least degrades the penalized criterion are combined, and the merged model is refitted with a few quick EM cycles (`quick_cycles`). A Dirichlet prior with parameter `a_dir` on the proportions π_k stabilizes the estimation of small classes during these merges.
3. **Tree.** The sequence of merges forms a deterministic tree, from K_0 classes down to a single class; for each value of K examined, it provides an initial partition.
4. **Final fit.** For each value of K retained, the model is fitted with full EM cycles (`full_cycles`, with `inner_cap` inner iterations per cycle) until the tolerance `tol` is reached. The procedure is repeated at most `n_outer` times (outer iterations), each time restarting from the best solution obtained.

The partitions are nested: moving from K to $K+1$ classes generally splits one class into two.

5.6 Choosing the number of classes

When `K` is left empty, the model is fitted for $K = 1, \dots, K_{\max}$ and the value of K minimizing the BIC (Schwarz, 1978) or the ICL (Biernacki et al., 2000) is retained:

$$\mathrm{BIC}_K = -2\,\ell_K(\hat{\boldsymbol{\vartheta}}) + d_K \log N, \qquad \mathrm{ICL}_K = \mathrm{BIC}_K + 2\,\mathrm{E}_K, \quad \mathrm{E}_K = -\sum_{i=1}^{N}\sum_{k=1}^{K} \hat{\tau}_{ik} \log \hat{\tau}_{ik}, \tag{5}$$

where ℓ_K is the maximized log-likelihood, $d_K = (K-1) + \mathrm{edf} + q(q+1)/2 + 1 + \mathbb{1}_{\text{penalty}}$ is the effective number of parameters, edf being the sum of the effective degrees of freedom of the K classes, covariate coefficients included (plus the membership coefficients if `weights = "covariates"`), and E_K is the entropy of the classification.

Criterion	What it favors	When to prefer it
BIC	Goodness of fit, penalized for model complexity	To estimate the structure of the population as accurately as possible
ICL	In addition, *well-separated* classes: the entropy term penalizes patients with uncertain class membership	To obtain clinically distinct profiles and clear-cut assignments

A class whose posterior size falls below `min_size` is removed and the model is refitted. When `K` is fixed, no selection is performed; the tree provides the initial partition.

Number of classes

The criterion identifies the value of K best supported by the data; it does not establish that each class is a clinical entity. Two very similar classes, or a class containing only a few patients, call for critical scrutiny; a simpler, interpretable and stable solution is often preferable to a marginal gain in the criterion.

5.7 Covariates

Covariates can enter the model in two ways:

— **in the trajectory**: each class has its own coefficients $\boldsymbol{\gamma}_k$; clustering then bears on the shape of the trajectory *for equal covariate values*;

— **in class membership** (`weights = "covariates"`): the probability of belonging to each class depends on patient characteristics measured at baseline, through a multinomial logistic model, $\pi_k(\mathbf{w}_i) = \exp(\boldsymbol{\alpha}_k^\top \mathbf{w}_i) / \sum_l \exp(\boldsymbol{\alpha}_l^\top \mathbf{w}_i)$, with class C1 as the reference category.

5.8 Assignment of patients

The posterior probability of class membership is

$$\hat{\tau}_{ik} = \frac{\hat{\pi}_k\, \varphi_k(\mathbf{y}_i)}{\sum_{l=1}^{K} \hat{\pi}_l\, \varphi_l(\mathbf{y}_i)}. \tag{6}$$

Each patient is assigned to the class with the highest posterior probability (maximum a posteriori rule).

5.9 Parameters

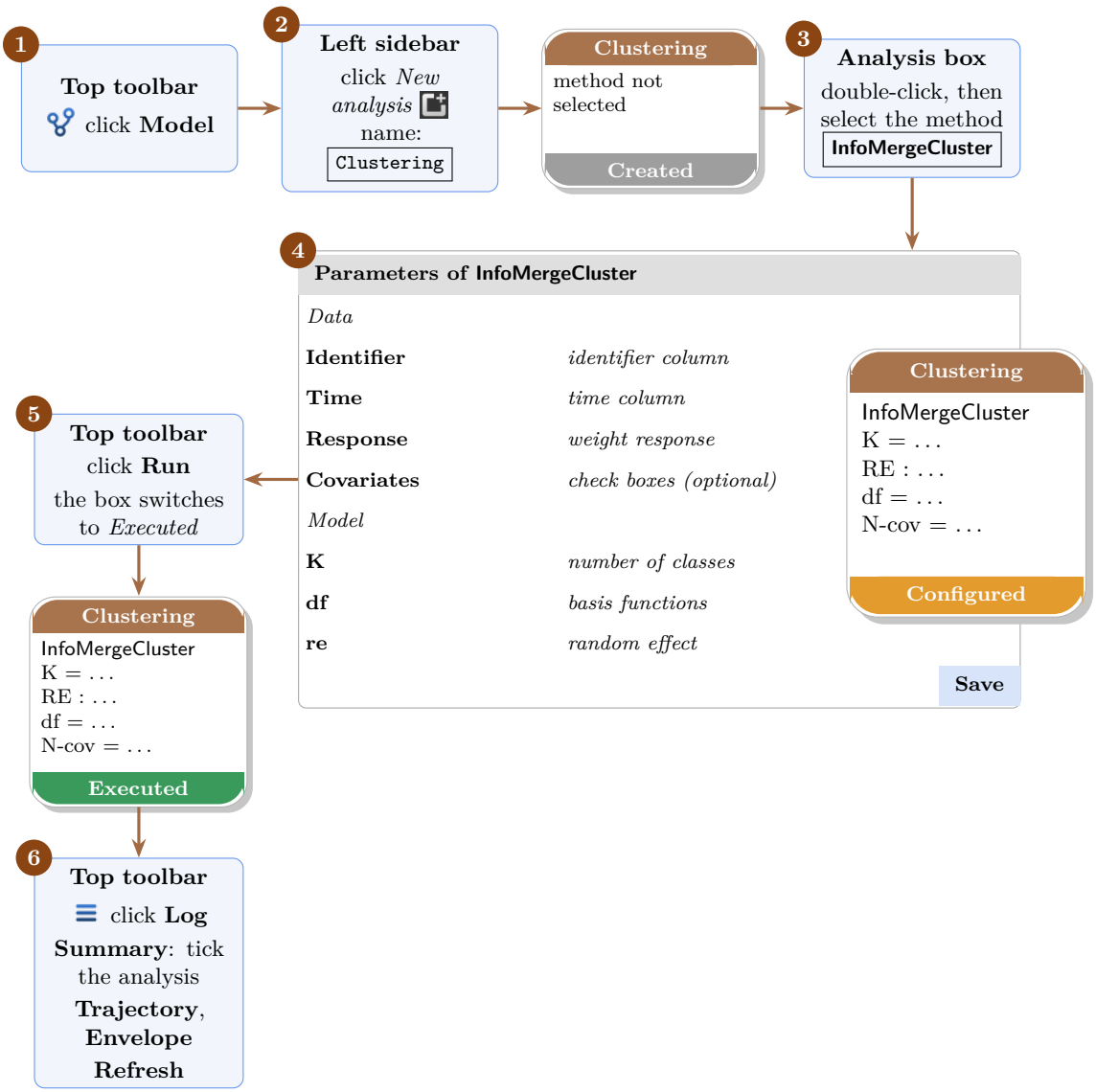
Parameter	Default	Role and advice
Model		
`K`	3	Number of classes; empty: automatic selection by the criterion
`K_max`	8	Maximum number of classes examined under automatic selection
`df`	15	Number of B-spline functions for the mean trajectory
`degree`	3	Spline degree (cubic)
`penalty`	ticked	Curvature penalty; should be kept
`pen_order`	2	Order of the penalized derivative
`re`	`linear`	Random-effect structure (Section 5.3)
`weights`	`constant`	`covariates`: covariate-dependent class membership
`criterion`	BIC	Criterion for selecting K: BIC or ICL
`min_size`	2	Minimum posterior size of a class
`K0`	empty	Number of micro-clusters at initialization; empty: automatic
Algorithm		
`a_dir`	0.5	Parameter of the Dirichlet prior used during merges
`n_outer`	3	Maximum number of outer iterations
`quick_cycles`, `full_cycles`	4, 15	Quick (exploration) and full (final fit) EM cycles
`inner_cap`	3	Inner iterations per cycle
`tol`	10^{-5}	Convergence tolerance
`nstart`	5	Number of k-means initializations
`seed`	empty	Random seed; fixing it makes the result reproducible
Plot		
`envelope`, `level`	no, 0.95	Confidence envelope around the mean trajectories (Section 8.2)
`show_data`, `max_subjects`	yes, 300	Individual trajectories and maximum number displayed
`xlab`, `ylab`, `legend`		Axis titles and class legend; the plot title is the box name (Section 8)

Reproducibility

The initialization is random (k-means): fixing `seed` makes the result reproducible. Checking that the classes are robust to a change of seed is a stability check that should be reported. The seed does not reproduce an R implementation: R uses the Mersenne Twister generator with `set.seed()`, whereas AbosCluster uses PCG64 (NumPy); the sampling functions (`sample`, `rmultinom`) and the k-means algorithms (Hartigan–Wong versus Lloyd with k-means++ seeding (Arthur and Vassilvitskii, 2007)) also differ. To compare the two, supply the same initial partition, or compare the final partitions using the adjusted Rand index (Section 8.3).

5.10 Clustering with InfoMergeCluster: a worked example

Two analyses are run with $K = 3$ imposed: body weight without covariates, and TWL with the four covariates (age, surgical procedure, diabetes status, smoking status). The diagram numbers the successive clicks; fields not mentioned keep their default values (Section 5.9).



Values entered at step 4 (columns `Patient` and `Time` for the identifier and time; $K = 3$, `df` $= 20$ and a `linear` random effect in both cases):

Name entered	Response	Covariates	re
Poids-IM	`Poids`	none	`linear`
TWL-IM	`TWL`	age, Technique, smoking, diabetes	`linear`

The covariates ticked are the columns `AgeJourIntervention__lcn` (age, quantitative), `ABOS_Intervention` (surgical procedure, three categories), `M000..TabacFumeur` (smoking status) and `M000..ABOS_Diabete_Oui__lc` (diabetes status, three categories), encoded as indicator variables (first category as reference). A single click on **Run** executes both analyses.

Displaying the results (step 6). **Log** opens the results (Section 8): tick both analyses under **Summary**, then **Trajectory** and **Envelope** (95%), and click **Refresh**; **Save plot** saves the plots.

Results: Table 6 and Figure 11.

Analysis	C1	C2	C3
Poids-IM : no-cov	58%	29%	12%
TWL-IM : with-cov	46%	30%	24%

Table 6 – Class proportions estimated by InfoMergeCluster ($K = 3$) for body weight and TWL (Figure 11), rounded to the nearest percent. Within each analysis, classes are numbered by decreasing proportion: C1 for weight and C1 for TWL do not necessarily comprise the same patients.

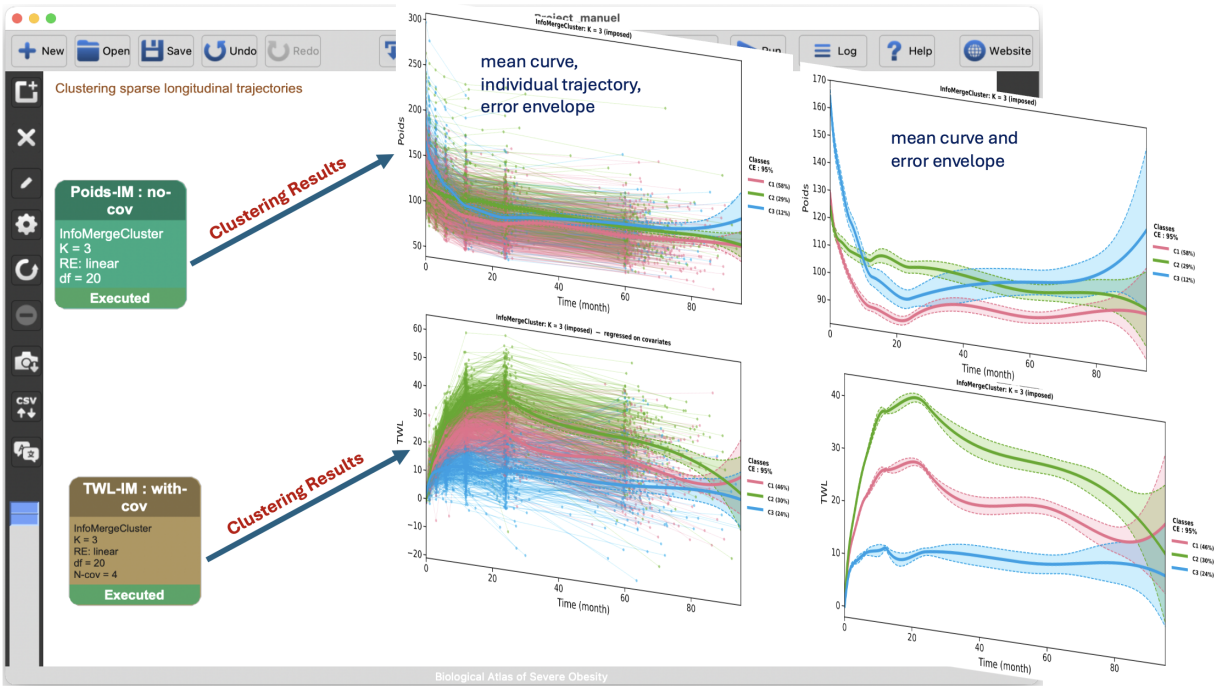


Figure 11 – InfoMergeCluster with $K = 3$ imposed, `linear` random effect and `df` $= 20$: analysis "Poids-IM : no-cov" (top) and analysis "TWL-IM : with-cov", adjusted for age at surgery, surgical procedure, smoking status and diabetes status (bottom). Left: individual trajectories, mean trajectories and 95% confidence envelopes; right: mean trajectories and envelopes only.

Poids-IM : no-cov. The classes differ both in initial weight and in their course over time.

— **C1 (58%)**, initial weight of about 120 to 130 kg: decrease to a nadir of the mean trajectory of about 83 kg at around 22 months, followed by stabilization between 84 and 88 kg until the end of follow-up.

— **C2 (29%)**, initial weight similar to that of C1: smaller weight loss, down to about 104 kg at around 12 months, then a slow decline, to about 100 kg at 30 months and 92 kg at 80 months; the gap with C1 thus reaches about 20 kg within the first year.

— **C3 (12%)**, highest initial weight (about 165 kg): the fastest and largest weight loss, about 75 kg over 22 months, down to a nadir close to 90 kg, followed by progressive weight regain (about 96 kg between 50 and 70 months). Its trajectory crosses that of C2 at around 10 months and again at around 45 months.

The crossing of C2 and C3 shows that the partition cannot be reduced to initial weight: the shape of the trajectory is what determines class membership, owing to the random intercept and slope (Section 7).

TWL-IM : with-cov. The classes differ in the magnitude and durability of weight loss.

— **C2 (30%)**, *good responders*: the greatest weight loss, about 38% by 10 months and a maximum of about 41% at around 20 months; then a decline, to about 34% at 40 months and 33% at 60 months, which becomes more pronounced beyond 70 months (about 28% at 80 months).

— **C1 (46%)**, *intermediate responders*: about 27% at 12 months and a maximum of about 29% at around 20 months, followed by partial loss of the benefit, with a plateau around 23–24% between 40 and 60 months and a decrease toward 20% at 80 months.

— **C3 (24%)**, *insufficient response*: TWL of about 11% by 10 months, then a plateau around 13% between 25 and 60 months, and a slight increase to 15–16% thereafter, within a wide envelope.

Trajectories are shown on raw data, at each class's mean covariate values; the comparison with regressed data is given in Figure 13.

The envelopes, narrow up to about 60 months, widen thereafter: beyond 70 months few patients are still being seen, and late changes (weight regain in C3 for body weight, decline of C2 for TWL) are imprecisely estimated. The two analyses do not define the same classes; their agreement is quantified in Section 8.3.

6. FKMS: functional k-means

FKMS (*functional k-means*) seeks the partition that brings each trajectory as close as possible to the mean curve of its class.

6.1 Principle

The mean curve of class k is $\mu_k(t) = \mathbf{S}(t)^\top \boldsymbol{\beta}_k$, where $\mathbf{S}(t)$ is a basis of `N.bs` functions (cubic B-splines or a Fourier basis). FKMS minimizes

$$\sum_{i=1}^{N} \sum_{j=1}^{n_i} w_{ij} \left(y_{ij} - \mathbf{S}(t_{ij})^\top \boldsymbol{\beta}_{c_i} - \mathbf{x}_{ij}^\top \boldsymbol{\gamma}_{c_i} \right)^2, \qquad w_{ij} = \frac{1}{n_i}, \tag{7}$$

by alternating, as in Lloyd's algorithm (Lloyd, 1982), the following two steps until convergence:

1. **estimation**: for a given partition, the coefficients of each class are obtained by penalized weighted least squares, $\hat{\boldsymbol{\theta}}_k = (\mathbf{D}_k^\top \mathbf{W}_k \mathbf{D}_k + \lambda_k \mathbf{P})^{-1} \mathbf{D}_k^\top \mathbf{W}_k \mathbf{y}_k$;
2. **assignment**: each patient is assigned to the class whose curve minimizes the sum of their squared deviations.

The weight $1/n_i$ gives every patient the same influence. Among `N.random` random starts, the solution with the smallest mean squared error (MSE) is retained.

6.2 Smoothing and selection of λ by generalized cross-validation

With a B-spline basis, curvature is penalized as in Section 5.2. Ticking `lambda.gcv` opens **Grid of lambda values (GCV)**: **Lower bound**, **Upper bound** and **Number of values** define a log-spaced grid (by default, 30 values from 10^{-6} to 10^{3}). The value of λ minimizing the generalized

cross-validation criterion (Craven and Wahba, 1979) is retained:

$$\mathrm{GCV}(\lambda) = \frac{n\,\mathrm{RSS}_w(\lambda)}{(n - \operatorname{tr}\mathbf{H}_\lambda)^2}, \tag{8}$$

where RSS_w is the weighted residual sum of squares and $\operatorname{tr}\mathbf{H}_\lambda$ the effective number of parameters. The smoothing parameter λ is class-specific, or common to all classes if `lambda.common` is ticked; without `lambda.gcv`, the value `lambda` is applied as is. Cancelling the grid dialog box unticks `lambda.gcv`; unticking and re-ticking it allows the grid to be modified. The grid is recalled in the analysis box (for example `grid: [100, 1000]` × `10`).

Choosing the λ grid

The optimal λ depends on the time unit, on `N.bs` and on the scale of the response (kg or %). If it falls on a boundary of the grid, widen the grid and rerun the analysis. The λ of each class is reported in the export (Section 8.4).

6.3 Covariates, confidence envelopes and number of classes

— `cov.effect = "cluster"`: class-specific covariate effects; `"common"`: a common effect during the search for the partition, followed by a class-specific final fit.

— `conf.band`: computes the covariance matrix of each class's coefficients, required for the confidence envelope (Section 8).

— `se.type`: estimator of this covariance matrix, `sandwich` or `model` (see below).

— `K` left empty: FKMS is run for $K = 1, \ldots, K_{\max}$ and the value of K with the smallest MSE is retained; this automatic selection is not recommended (see the note below).

Covariance estimator (`se.type`). Let $\mathbf{A}_k = \mathbf{D}_k^\top \mathbf{W}_k \mathbf{D}_k + \lambda_k \mathbf{P}$ and let $\hat{r}_{ij}$ denote the residuals of class k. The `sandwich` estimator (Liang and Zeger, 1986), used by default, aggregates the score contributions by patient:

$$\hat{\mathbf{V}}_k = \mathbf{A}_k^{-1}\Big(\tfrac{N_k}{N_k-1}\sum_{i\in k}\mathbf{U}_i\mathbf{U}_i^\top\Big)\mathbf{A}_k^{-1}, \qquad \mathbf{U}_i = \sum_{j=1}^{n_i} w_{ij}\,\hat{r}_{ij}\,\mathbf{d}_{ij}, \tag{9}$$

where $\mathbf{d}_{ij}$ is the row of the design matrix (basis functions and covariates) for visit j of patient i, and N_k the number of patients in class k. It remains valid whatever the within-patient correlation structure. The `model` estimator assumes independent, homoscedastic errors, $\hat{\mathbf{V}}_k = \hat{\sigma}_k^2\,\mathbf{A}_k^{-1}\mathbf{D}_k^\top\mathbf{W}_k^2\mathbf{D}_k\mathbf{A}_k^{-1}$; with repeated measurements, it underestimates the uncertainty.

MSE as a criterion for choosing K

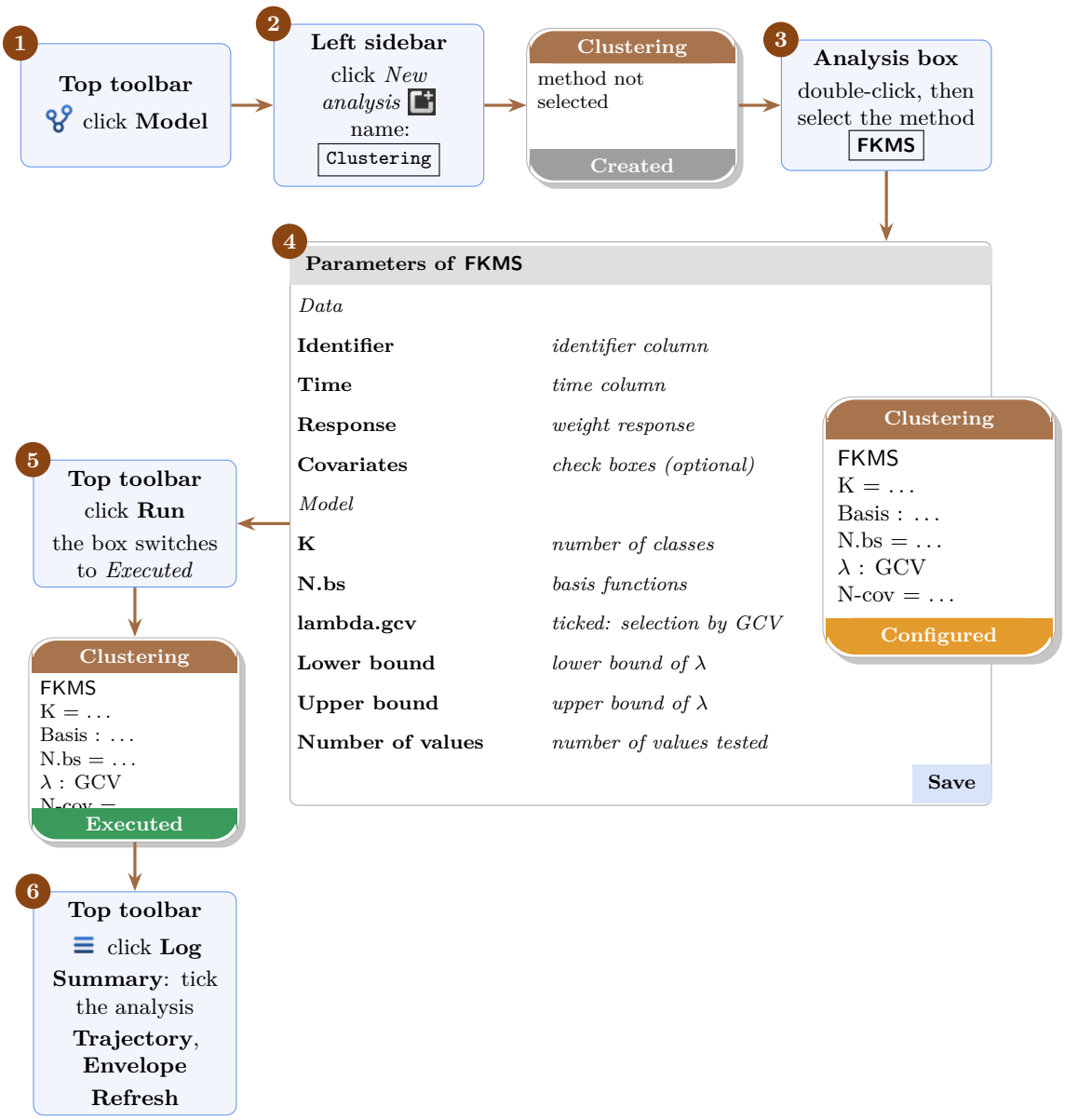
The MSE decreases as K increases: the "K with minimum MSE" is almost always $K_{\max}$, which amounts to no selection at all. Set K on clinical grounds, or use the value selected by InfoMergeCluster (BIC or ICL), and then compare the partitions (Section 8.3). The "elbow" of the MSE-versus-K curve is only an informal guide.

Parameter	Default	Role
`K`, `K_max`	3, 8	Number of classes (empty: minimum MSE, i.e. in practice $K_{\max}$; not recommended); upper bound for automatic selection

Parameter	Default	Role
`N.bs`	8	Number of basis functions (at least 4 for B-splines, 3 for Fourier)
`basis`	B	`B`: cubic B-splines; `F`: Fourier basis
`lambda`, `lambda.gcv`, `lambda.common`	0, yes, no	Fixed smoothing, GCV-selected smoothing, class-specific or common λ
λ grid (**Lower bound**, **Upper bound**, **Number of values**)	10^{-6}, 10^{3}, 30	Values of λ evaluated by GCV, log-spaced; entered when `lambda.gcv` is ticked
`cov.effect`	`cluster`	Class-specific or common covariate effects
`conf.band`, `se.type`	yes, `sandwich`	Computation of the coefficient covariance and choice of estimator (pointwise confidence envelope)
`N.random`, `N.rep`, `seed`	10, 100, empty	Random initializations, maximum number of iterations, seed

6.4 Clustering with FKMS: a worked example

The same analyses as in Section 5.10 are run with FKMS: "Poids-FK" (no covariates) and "TWL-FK" (four covariates), with $K = 3$, $N.bs = 20$ B-splines and λ selected by GCV over 10 values between 100 and 1000. Fields not mentioned keep their default values (Section 6.3).



The grid fields appear only once `lambda.gcv` is ticked. `basis` = `B`. Display settings are the same as in Section 5.10.

Results: Table 8 and Figure 12.

Analysis	C1	C2	C3
Poids-FK : no-cov	46%	42%	12%
TWL-FK : with-cov	43%	31%	26%

Table 8 – Class proportions estimated by FKMS ($K = 3$) for body weight and TWL (Figure 12), rounded to the nearest percent. Within each analysis, classes are numbered by decreasing size.

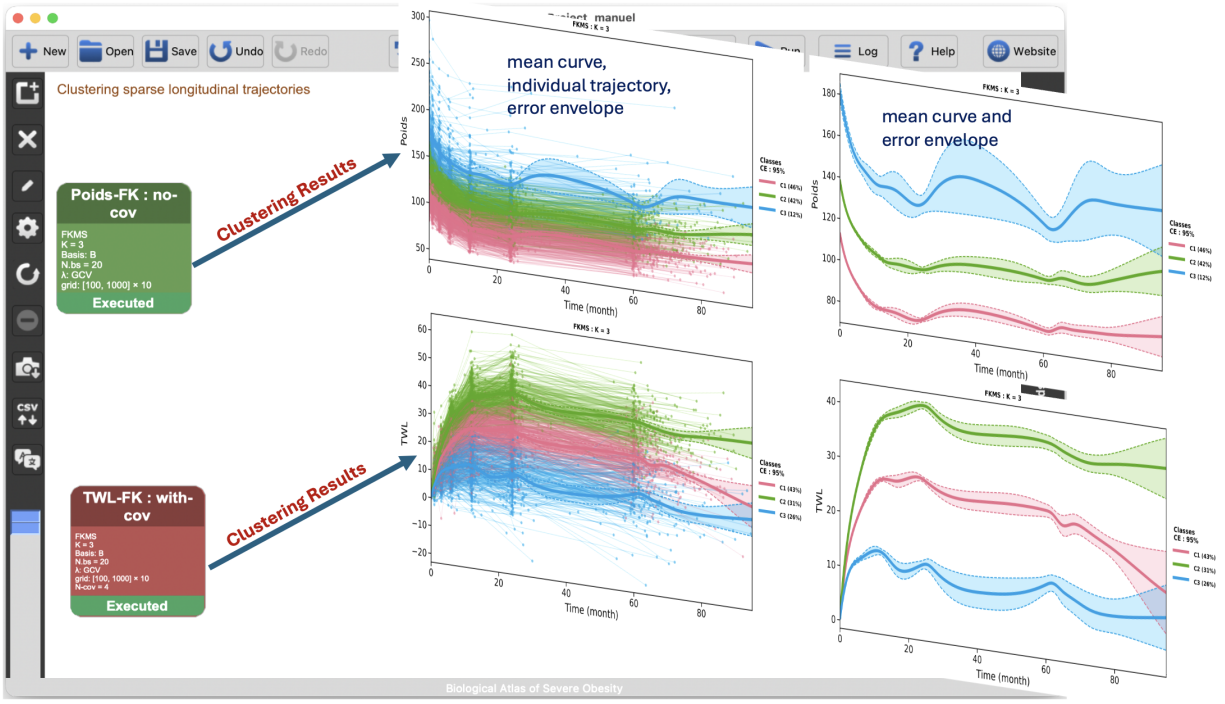


Figure 12 – FKMS with $K = 3$ imposed: analysis "Poids-FK : no-cov" (top) and analysis "TWL-FK : with-cov", adjusted for age at surgery, surgical procedure, smoking status and diabetes status (bottom), both with $N.bs = 20$ and λ selected by GCV over a grid of 10 log-spaced values between 100 and 1000. Left: individual trajectories, mean trajectories and 95% pointwise confidence envelopes (`sandwich` estimator); right: mean trajectories and envelopes only.

Poids-FK : no-cov. The classes are ordered by initial weight and do not cross.

— **C1 (46%)**, lowest initial weight (about 112 kg): decrease to about 76 kg at 12 months and 71 kg at around 22 months, then a slow decline, to around 65 kg beyond 60 months.

— **C2 (42%)**, intermediate initial weight (about 138 kg): decrease to about 100 kg at 12 months, then values between 88 and 97 kg until the end of follow-up.

— **C3 (12%)**, highest initial weight (about 180 kg): decrease to about 127 kg at around 22 months, followed by an irregular trajectory between 115 and 140 kg. The envelope, wide from 25 months onward, shows that these fluctuations are not established (small class, few late visits).

TWL-FK : with-cov. The same hierarchy as with InfoMergeCluster is found.

— **C2 (31%)**, *good responders*: about 37% at 12 months, a maximum of about 39% between 20 and 25 months, then a slow decline, to about 34% at 40 months, 32% at 60 months and 28% beyond 75 months.

— **C1 (43%)**, *intermediate responders*: about 26% between 12 and 24 months, then a progressive decrease, to about 21% between 40 and 60 months, more marked beyond 65 months.

— **C3 (26%)**, *insufficient response*: a maximum of about 12% at around 10 months, then a decline, to about 5% between 40 and 55 months and close to 0% beyond 75 months, i.e. a return to preoperative weight.

The envelopes widen at the end of follow-up: caution is warranted beyond 70 months.

Comparison with InfoMergeCluster. For TWL, the same three profiles are found, with similar proportions (43, 31 and 26% with FKMS; 46, 30 and 24% with InfoMergeCluster). The mean trajectories agree up to about 24 months. Thereafter, good responders remain close (about 33% at 60 months with InfoMergeCluster and 32% with FKMS), as do intermediate responders (about 23–24% versus 21% between 40 and 60 months). The difference concerns the insufficient-response class: it remains around 13% with InfoMergeCluster, but falls to about 5% between 40 and 55 months with FKMS, and then toward 0% beyond 75 months. This discrepancy appears as early as 40 months, while the InfoMergeCluster envelope is still narrow; it cannot, therefore, be attributed solely to the imprecision of late follow-up. Beyond 70 months, however, the envelopes widen and between-method differences are difficult to interpret. For body weight, the partitions diverge (46, 42 and 12% versus 58, 29 and 12%). FKMS orders the classes by initial weight, whereas InfoMergeCluster separates two classes with similar initial weight according to the magnitude of weight loss. Lacking a random effect, FKMS groups patients by level whenever level dominates the variability, as is the case for absolute weight (Section 7). Similar proportions do not prove that the same patients are grouped together: see the ARI and the Jaccard index (Section 8.3).

7. InfoMergeCluster vs FKMS

Table 9 summarizes the differences between the two methods.

Table 9 – Characteristics of the two methods, as implemented in this study.

	InfoMergeCluster	FKMS
Family	mixture model (penalized likelihood)	partitioning (k-means type)
Class mean trajectory	penalized B-splines, λ estimated by the Fellner–Schall method	B-splines, penalized weighted least squares, λ selected by GCV
Basis dimension	`df`; knots at quantiles of all measurement times, coincident knots removed (effective dimension $\leq$ `df`)	`N.bs`; knots at quantiles of the distinct measurement times, without removal (dimension equal to `N.bs`)
Individual variability	random effect (intercept, linear or spline)	not modeled
Measurement error	σ^2 estimated (prior calibrated by variogram)	implicit (squared-error loss)
Covariates	class-specific, unpenalized effects $\boldsymbol{\gamma}_k$	class-specific, unpenalized effects $\boldsymbol{\gamma}_k$
Assignment	probabilistic (τ_{ik}), then class of maximum posterior probability	deterministic (nearest center)
Number of classes	imposed, or selected by BIC or ICL	imposed, or selected by minimum MSE (not recommended, Section 6.3)
Initialization	k-means micro-clusters (random, `seed`), then deterministic merge tree	random assignments, best partition retained
Confidence envelope	pointwise; covariance $\mathbf{H}_k^{-1}$ derived from the model (penalized Fisher information)	pointwise; patient-level `sandwich` covariance (default) or `model`

Key difference. Both methods estimate one mean trajectory per class, but they handle within-class variability differently. InfoMergeCluster models it explicitly (random effect $\boldsymbol{u}_i$, residual error $\boldsymbol{\varepsilon}_i$): class assignment depends mainly on the *shape* of the trajectory. FKMS treats it as squared

error: any difference in level carries weight, and patients are grouped by level whenever level dominates.

8. Reading, comparing and exporting results

8.1 The results panel

After **Run**, or through **Log**, one plot is displayed per executed analysis, showing the individual trajectories colored by class, the mean trajectories and a legend ("C1 (46%)"). The title is the box name followed by `no-cov` or `with-cov`.

The **Summary** list selects the analyses displayed:

— **All analyses**, ticked by default, displays every analysis;

— the other check boxes identify each analysis by method and response ("InfoMergeCluster : TWL"), as do the columns **Model 1** and **Model 2** of the comparison results; two analyses with the same method and response are distinguished by their box name, added in parentheses;

— ticking an analysis displays it alone, ticking further analyses adds them, and unticking the last one reverts to **All analyses**.

When at least one comparison box exists, the **Comparisons** check box displays the agreement indices (Section 8.3).

8.2 Plot-specific options

These options apply to the selected plot and are stored for each analysis. The plot is redrawn without re-estimation, at the same scroll position.

Option	Effect
Trajectory	Displays the individual trajectories; **All** shows them all, **TN** sets their number (reproducible random draw)
Envelope	Displays the envelope of the mean trajectories; the legend states its level, "CE : 95%"
Regressed data	For a covariate-adjusted model: data and trajectories regressed on the covariates (see below)
Comparisons	Present if a comparison box exists: displays, in a movable box, the adjusted Rand index and the Jaccard index for each pair of compared analyses

Confidence envelope. Both methods produce a *pointwise* confidence envelope (CE) for the *mean trajectory*:

$$\mathrm{CE} : \hat{\mu}_k(t) \pm z_{1-\alpha/2} \sqrt{\mathbf{a}(t)^\top \hat{\mathbf{V}}_k \, \mathbf{a}(t)}, \qquad 1-\alpha = \texttt{level}, \tag{10}$$

where $\mathbf{a}(t)$ gathers the basis functions evaluated at time t and, in covariate-adjusted models, the covariate values used for display (see below), $z_{1-\alpha/2}$ is the quantile of the standard normal distribution (1.96 for $\texttt{level} = 0.95$), and $\hat{\mathbf{V}}_k$ is the estimated covariance matrix of the coefficients, which is the only element that differs between the two methods:

— **InfoMergeCluster**: $\hat{\mathbf{V}}_k = \mathbf{H}_k^{-1}$, with $\mathbf{H}_k = \sum_i \hat{\tau}_{ik}\, \mathbf{X}_i^\top \mathbf{V}_i^{-1} \mathbf{X}_i + \hat{\lambda}\mathbf{P}$ and $\mathbf{V}_i = \mathbf{Z}_i \hat{\mathbf{G}} \mathbf{Z}_i^\top + \hat{\sigma}^2 \mathbf{I}_{n_i}$: the penalized Fisher information, weighted by the posterior membership probabilities. The random effect accounts for within-patient correlation.

— **FKMS**: patient-level `sandwich` estimator (default) or `model` estimator (Section 6.3).

At the 95% level, the envelope covers the mean trajectory at each time point taken separately, not simultaneously over the whole follow-up period.

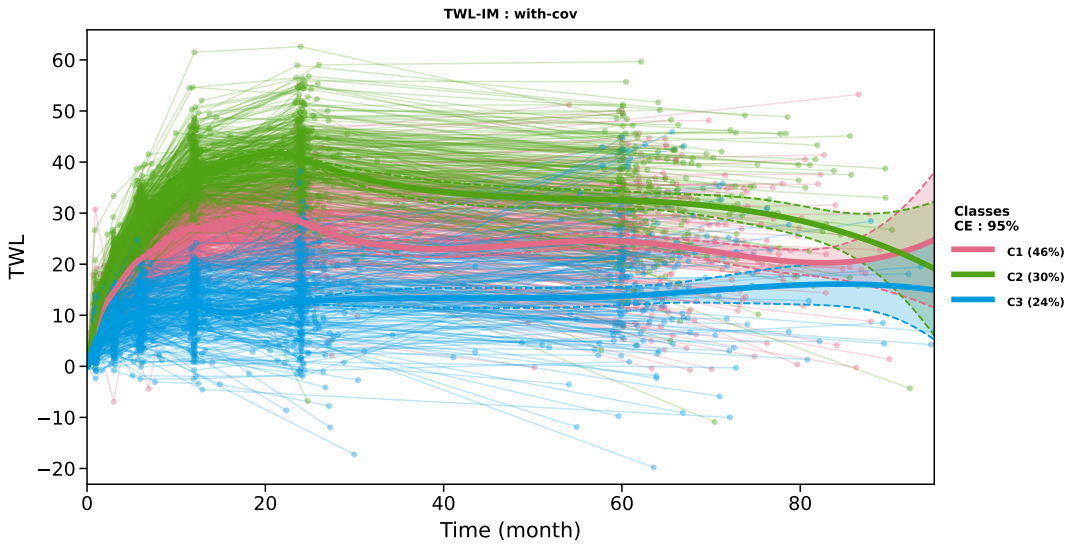
Reading a confidence envelope

Two envelopes that do not overlap over a given period indicate clearly different mean trajectories. The envelope pertains to the mean curve, not to the dispersion of individual patients. It is *conditional* (partition, variance components and smoothing treated as known, assignment uncertainty ignored) and is therefore probably too narrow, especially for small classes.

Raw data and regressed data. In covariate-adjusted models, two representations are available:

— **raw data**: weight measurements as observed; the trajectory of each class is computed at the mean covariate values *of that class*, $\mathbf{B}(t)^\top \hat{\boldsymbol{\beta}}_k + \bar{\mathbf{x}}_k^\top \hat{\boldsymbol{\gamma}}_k$, and runs through the middle of its patients' trajectories;

— **regressed data**: the covariate effect is removed, $y_{ij}^{\text{reg}} = y_{ij} - (\mathbf{x}_{ij} - \bar{\mathbf{x}})^\top \hat{\boldsymbol{\gamma}}_{c_i}$, which brings every patient to the mean covariate values of the whole sample; the trajectories are recomputed accordingly.



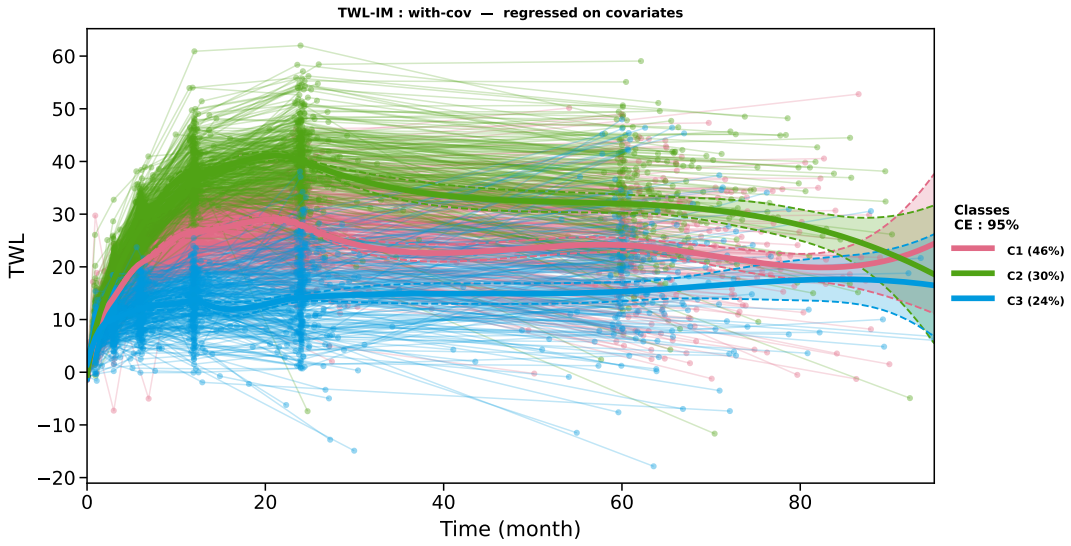


Figure 13 – Analysis "TWL-IM : with-cov" (InfoMergeCluster, $K = 3$ imposed), adjusted for age at surgery, surgical procedure, smoking status and diabetes status: raw data (top) and data regressed on these four covariates (bottom). The legend shows the proportion of each class and the level of the confidence envelope (CE : 95%).

The partition is identical; only the ordinates change. For class k, the difference between raw and regressed trajectories equals $(\bar{\mathbf{x}}_k - \bar{\mathbf{x}})^\top \hat{\boldsymbol{\gamma}}_k$: the effect on TWL of the composition of the class (age, surgical procedure, smoking status, diabetes status) relative to the whole sample.

Regressed data compare classes "at equal profile"; raw data describe what patients actually experienced. In both cases, the $\hat{\boldsymbol{\gamma}}_k$, estimated for a fixed partition, are not causal effects.

8.3 Comparing several analyses

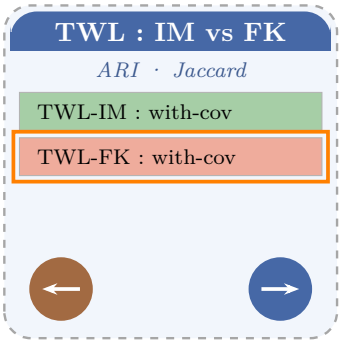
Analyses within the same project can be compared in two ways:

— **Visually**: display the analyses together, with the same axis limits (**Zoom axes** with no plot selected).

— **Quantitatively**: patient-by-patient agreement (ARI and Jaccard index), computed in *model comparison boxes*.

Comparison boxes

As soon as a comparison box exists (*New comparison* , right sidebar), the panel is split: analysis boxes on the left (55%) and comparison boxes on the right (45%, titled **Comparisons**).



Comparison name (optional color)
Indices selected in **Settings**

One row per analysis: title and color of its box;
orange frame: row selected for removal;
blue frame: analysis box selected on the left

Left arrow: remove the selected analyses;
right arrow: add the selected analysis boxes

Figure 14 – A comparison box containing two analyses. Each arrow appears only when it can be used.

1. **Create the box**: click *New comparison*, enter a name (for example "TWL : IM vs FK"), then choose a color or keep the default one. The initial size is 80%, adjustable with the slider of the right sidebar.
2. **Add analyses**: select one or more analysis boxes (Shift, Ctrl or Cmd + click to select several), then click the *right arrow* of the comparison box. Duplicates are not allowed; each analysis occupies one row, shown in the color of its box and bearing its title (box name followed by `no-cov` or `with-cov`). The **Settings** list of the comparison also shows the method and response of each analysis.
3. **Remove analyses**: click one or more rows of the comparison box (orange frame, other rows dimmed), then the *left arrow*. An analysis box selected on the left is framed in blue in every comparison that contains it and can be removed in the same way.
4. **Choose the indices**: *Configure* (right sidebar) or double-click the box. Tick **ARI** (default) and/or **Jaccard**, located side by side under **Comparison index**; set the color (**Choose...**, **Default**), then click **Save**.
5. **Read the indices**: after **Run**, tick **Comparisons** in the results panel.

Deletion, renaming and restoration are performed from the right sidebar; comparisons follow any changes made to the analysis boxes. All actions can be undone with **Undo**.

Agreement indices

Let $\mathcal{P}$ and $\mathcal{Q}$ be two partitions of the same M patients, n_{uv} the number of patients assigned to class u of $\mathcal{P}$ and to class v of $\mathcal{Q}$, and $a_u = \sum_v n_{uv}$ and $b_v = \sum_u n_{uv}$ the class sizes. Both indices rely on counting *pairs* of patients:

$$S_{11} = \sum_{u,v} \binom{n_{uv}}{2}, \qquad S_a = \sum_u \binom{a_u}{2}, \qquad S_b = \sum_v \binom{b_v}{2}, \tag{11}$$

where S_{11} counts the pairs placed together in both partitions, and S_a and S_b the pairs placed together in $\mathcal{P}$ and in $\mathcal{Q}$, respectively. The adjusted Rand index (Hubert and Arabie, 1985) and the Jaccard index (Jaccard, 1912) are given by

$$\mathrm{ARI} = \frac{S_{11} - S_a S_b / \binom{M}{2}}{\frac{1}{2}(S_a + S_b) - S_a S_b / \binom{M}{2}}, \qquad J = \frac{S_{11}}{S_a + S_b - S_{11}}. \tag{12}$$

Index	Range	Interpretation
ARI	≤ 1; expected value 0 under independence	Chance-corrected agreement: 1 for identical partitions, close to 0 for the agreement expected by chance given the class sizes, negative for agreement below chance
Jaccard	$[0, 1]$	Proportion of pairs placed together by both partitions among the pairs placed together by at least one of them; not corrected for chance, it increases mechanically when classes are few and large

The indices are invariant to class labels (C1 in one analysis may correspond to C2 in another). They are computed for each pair of executed analyses, over the M patients classified by both. Here, since rows with missing smoking status were removed beforehand (Appendix, Section A.2), $M = 1\,438$ for every pair. Without this cleaning step, a covariate-adjusted analysis excludes patients with incomplete covariates, and M is reduced to the patients common to both analyses.

Displaying and exporting the indices

Comparisons opens a movable *Comparison results* box, with one section per comparison and one row per pair: **Model 1**, **Model 2** (method, response and number of classes retained; box name in parentheses when two analyses share the same method and response), **N** (number of patients M), **ARI** and **Jaccard**. Indices are rounded to three decimal places; an index that was not selected is shown as "—". A message flags incomplete comparisons. **Export CSV** saves the indices at full precision, with analyses identified by their box name; **Close** closes the box.

Example: three analyses of body weight

This example applies the preceding steps to three analyses of body weight run with $K = 3$ (Figure 15; Sections 5.10 and 6.4): InfoMergeCluster without covariates ("Poids-IM"), InfoMergeCluster with the four covariates ("Poids-IM2") and FKMS without covariates ("Poids-FK"). Since two boxes cannot share the same name, the second is named "Poids-IM2". Each box has its own color, which is carried over to the comparison boxes; the blue outline indicates selection.

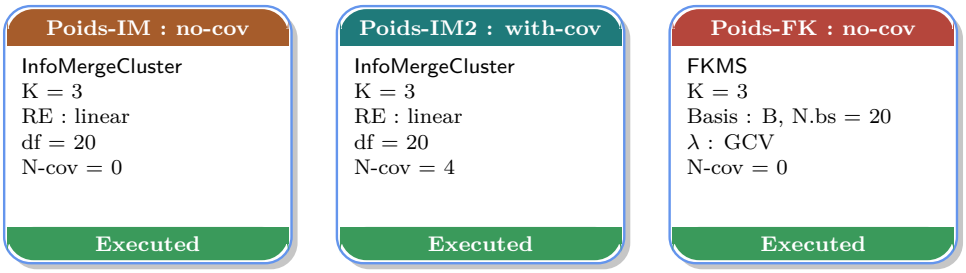


Figure 15 – Three executed analysis boxes, selected (blue outline) in order to be placed in the same comparison box.

The pair "Poids-IM" / "Poids-FK" measures the effect of the method, for the same response and covariates; the pair "Poids-IM" / "Poids-IM2" measures the effect of covariate adjustment, for the same method.

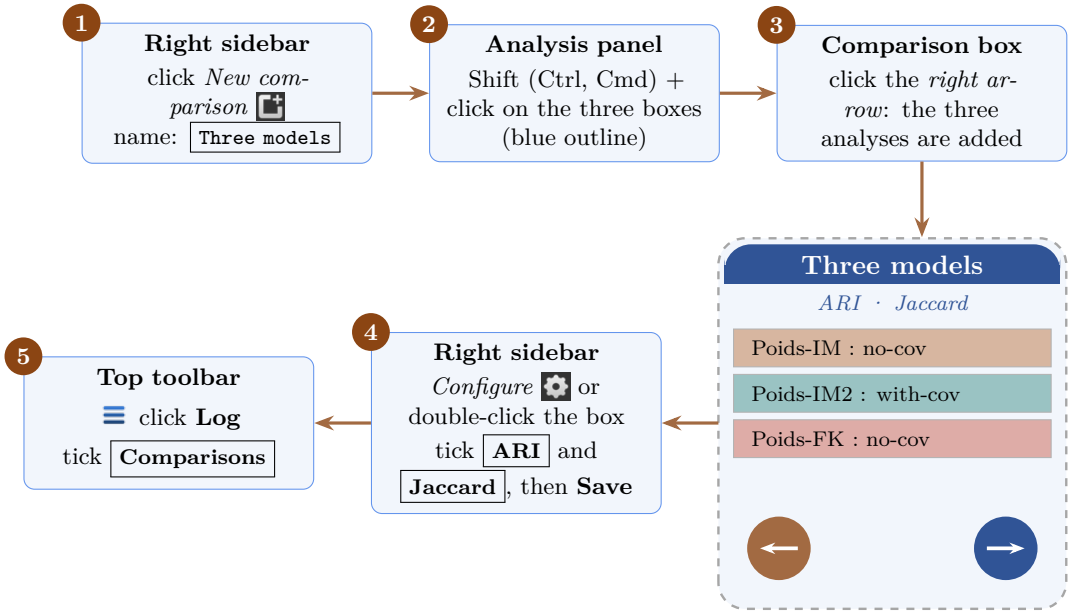


Figure 16 – Steps for comparing the three analyses (same steps as in the list above). At step 3, the comparison box receives one row per analysis, shown in the color of its box and identified by that box's title (name followed by `no-cov` or `with-cov`).

The indices are computed from the stored partitions, without rerunning **Run**; any analysis that is not Executed must be run before step 5. Three analyses yield three pairs, all based on $M = 1\,438$ patients (Table 10).

Model 1	Model 2	N	ARI	Jaccard
InfoMergeCluster : Poids (Poids-IM)	InfoMergeCluster : Poids (Poids-IM2)	1 438	0.942	0.937
InfoMergeCluster : Poids (Poids-IM)	FKMS : Poids	1 438	0.063	0.299
InfoMergeCluster : Poids (Poids-IM2)	FKMS : Poids	1 438	0.060	0.299

Table 10 – *Comparison results* box for the "Three models" comparison ($K = 3$ for all three analyses): one row per pair of analyses, over the 1 438 patients classified by both analyses. Indices rounded to three decimal places.

Interpretation. The indices are read as described above (agreement indices). Covariate adjustment changes the InfoMergeCluster partition of body weight only slightly: ARI of 0.94 and Jaccard index of 0.94 between "Poids-IM" and "Poids-IM2". Conversely, the InfoMergeCluster and FKMS partitions are almost unrelated: an ARI of about 0.06 is barely above chance-level agreement. The Jaccard index of about 0.30 is not contradictory: since it is not corrected for chance, it remains high whenever classes are large. This disagreement is the one described in Section 6.4: FKMS groups patients according to their initial weight level, whereas InfoMergeCluster groups them according to the shape of their trajectory (Section 7).

Uses of comparison

— **Methods**: same response, two methods ("Poids-IM" versus "Poids-FK", ARI 0.06 in the example). A low ARI despite similar proportions indicates classes of similar size but composed of different patients.

— **Adjustment**: the same analysis without and with covariates ("Poids-IM" versus "Poids-IM2", ARI 0.94).

— **Responses**: body weight versus TWL, i.e. absolute weight level versus relative weight loss.

— **Stability**: the same configuration run with two different seeds; to be reported together with the results.

8.4 Exporting

— **Export of analyses** CSV: a CSV file describes each analysis (method, status, response, covariates, main parameters, retained K, proportions, BIC, ICL and log-likelihood for InfoMergeCluster, MSE and per-class λ for FKMS, numbers of patients and measurements). The file `<name>_clusters.csv` gives each patient's class for every analysis, providing the basis for post hoc analyses.

— **CSV export of comparisons** (*Comparison results* box): one row per pair of compared analyses, with the comparison name, the two analyses (identified by their box name), the number of patients, and the ARI and Jaccard index at full precision.

— **Save plot** : plots in PNG, PDF, SVG, JPEG or TIFF format at the chosen resolution, or gathered in a Word document; each plot's title is the name of its analysis box.

— **Screenshot** : image of the window.

— **Save**: data, parameters and results, allowing the study to be reproduced.

A. Description of the ABOS cohort data

The three longitudinal responses (body weight, BMI, TWL) are all derived from body weight.

A.1 Data and observation scheme

The source sample comprises $N = 1\,443$ patients from the ABOS cohort, for a total of $n = \sum_{i=1}^{N} n_i = 8\,734$ observations, with no duplicated (patient, time) pair. The number of observations per patient, n_i, ranges from 2 to 7 (mean 6.05; standard deviation 1.07; median 6), and the measurement times t_{ij} range from 0 to 95 months after surgery. Only smoking status has missing values (Section A.2).

The data come from the ABOS cohort (*A Biological Atlas of Severe Obesity*, NCT01129297), a prospective single-center cohort of patients with severe or morbid obesity undergoing bariatric surgery at Lille University Hospital (CHU de Lille), with enrollment since 2006. The source files contain 16 variables (see Table 11), grouped into four categories:

— **Identification and observation scheme**: `Patient`, `Time`, `n_obs`, `Dataset`.

— **Repeated anthropometric measurements**: `Poids`, `BMI`, `TWL`.

— **Baseline values (at surgery, $t = 0$)**: `M000..Clini_Poids`, `M000..Clini_BMI`.

— **Time-invariant baseline covariates**: `IEX_PATIENT..ABOS_taillecm`, `AgeJourIntervention__lcn`, `ABOS_Intervention`, `M000..TabacFumeur`, `M000..ABOS_Diabete_Oui__lc`, `M000_AnneeAvecDiab`, `TWL_pre_surg`.

Table 11 – Data dictionary.

Variable	Description	Unit
`Patient`	Identifier of subject i, $i = 1, \ldots, N$; statistical unit	–
`Time`	Postoperative time t_{ij}	months
`TWL`	Total Weight Loss TWL_{ij} of subject i at time t_{ij}	%
`Poids`	Body weight P_{ij} of subject i at time t_{ij}	kg
`BMI`	Body mass index BMI_{ij} of subject i at time t_{ij}	$\mathrm{kg/m^2}$
`IEX_PATIENT..ABOS_taillecm`	Height h_i of subject i	cm
`ABOS_Intervention`	Surgical technique (gastric bypass, sleeve gastrectomy, gastric banding)	–
`M000..Clini_Poids`	Baseline weight P_{i0}, measured on the day of surgery	kg
`M000..Clini_BMI`	Baseline BMI BMI_{i0}, measured on the day of surgery	$\mathrm{kg/m^2}$
`AgeJourIntervention__lcn`	Age a_i at surgery, time-invariant	years
`M000..TabacFumeur`	Smoking status (smoker, non-smoker; missing for 5 subjects)	–
`TWL_pre_surg`	Preoperative weight loss	%
`M000..ABOS_Diabete_Oui__lc`	Diabetes status (0 = no diabetes, 1 = T1D, 2 = T2D)	–
`Dataset`	Data source	–
`M000_AnneeAvecDiab`	Duration of diabetes	years
`n_obs`	Number n_i of observations of subject i	–

Variables are listed in the order in which they appear in the source files (14 833 rows, all cohorts combined). The prefix M000 denotes a measurement collected at the inclusion visit (month 0, day of surgery). Most variables are stored as character strings in the source files and converted to numeric format during preprocessing. T1D and T2D denote type 1 and type 2 diabetes, respectively.

Cohort source: *A Biological Atlas of Severe Obesity (Biological Tissue Collection)*, NCT01129297, ClinicalTrials.gov, `https://clinicaltrials.gov/study/NCT01129297`.

A.2 Missing values

Missing values before exclusion, by postoperative time (Table 12).

Table 12 – Missing values by variable and postoperative time, before exclusion (source sample).

	Postoperative time (months)								
Variable	≤ 0	$(0, 6]$	$(6, 12]$	$(12, 24]$	$(24, 36]$	$(36, 60]$	> 60	Total	Patients
`M000..TabacFumeur`	5 (0.3)	11 (0.3)	5 (0.4)	2 (0.2)	2 (0.3)	1 (0.7)	–	26 (0.3)	5
Number of measurements	1 443	3 362	1 281	1 255	688	141	564	8 734	1 443

Number of measurements with a missing value for the variable; in parentheses, percentage of the measurements taken within the time interval that are missing (Total column: percentage of all measurements). Patients: number of patients with at least one missing value for the variable. Variables not listed (identifier, time, body weight, BMI, TWL, age, diabetes status, surgical procedure) have no missing values. Codes treated as missing: empty string, `NA`, `N/A`, `NaN`, `NULL`, `..`

— **Mechanism.** Smoking status is a baseline covariate, recorded at inclusion and constant within patient. It is missing for 5 patients (0.3% of patients), and therefore for all 26 of their measurements. The distribution of these missing values over time simply follows the visit schedules of these five patients: missingness occurs at the patient level, not at the visit level.

— **Handling.** The rows concerned are removed, leaving $N = 1\,438$ patients and $n = 8\,708$ observations (median 6 per patient) for all analyses. Given its size (0.3% of patients), this exclusion has little bearing on the partitions.

A.3 Deterministic relationships between the response variables

Let h_i be the height of subject i (variable `IEX_PATIENT..ABOS_taillecm`, converted to meters). The following relationships hold:

$$\mathrm{BMI}_{ij} = \frac{P_{ij}}{h_i^2}, \qquad \mathrm{BMI}_{i0} = \frac{P_{i0}}{h_i^2}, \tag{13}$$

$$\mathrm{TWL}_{ij} = 100\,\frac{P_{i0} - P_{ij}}{P_{i0}} = 100\left(1 - \frac{\mathrm{BMI}_{ij}}{\mathrm{BMI}_{i0}}\right). \tag{14}$$

The following properties, relevant to clustering, follow directly.

— **BMI and body weight.** A patient's BMI is their body weight multiplied by $1/h_i^2$: the profiles are proportional, with a patient-specific scale factor. This normalization removes the effect of height, but not that of initial body size.

— **TWL and body weight.** TWL, the recommended outcome measure for reporting weight loss after bariatric surgery (Corcelles et al., 2016), is an affine transformation of body weight with patient-specific coefficients. It equals zero at $t = 0$, is positive in case of weight loss and negative when weight exceeds baseline; it thus removes the initial level and measures the relative change in weight.

— **Dependence between the three analyses.** The three analyses are based on the same underlying signal and are not independent. For body weight and BMI, between-patient variance is driven mainly by the initial level; for TWL, by the dynamics of weight loss.

Means and standard deviations of the three responses are given in Table 13.

Table 13 – Marginal descriptive statistics of the three response variables (source sample, before exclusion).

Response variable	Unit	Mean (standard deviation), $n = 8\,734$
Body weight (`Poids`)	kg	109.8 (26.9)
BMI	kg/m^2	38.9 (8.5)
TWL	%	16.4 (12.9)

Computed over the 8 734 observations, all time points combined: orders of magnitude, not changes over time.

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